

1 Global export of biomass and contaminants from water 2 to land by aquatic insects

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39 SUMMARY

40 Animals redistribute energy and materials across ecosystems, but these flows can
41 carry both benefits and risks ¹⁻⁴. Emergence of adult aquatic insects provides critical
42 nutrient subsidies to terrestrial consumers such as birds, bats and spiders ^{3,5}. Yet, the
43 extent to which these aquatic subsidies also deliver contaminants back to land
44 globally remains unknown ^{1,6}.

45 We synthesized reported aquatic insect production and emergence (305 records
46 across all major biomes) and more than 13 million observations of waterborne
47 contaminants to generate the first worldwide estimates of aquatic insect biomass and
48 contaminant transfers from rivers to land. We estimate that flowing waters export
49 ~1,600 mg dry mass of adult insects m⁻² of surface water y⁻¹, equivalent to ~1.8 million
50 tons of dry mass and ~17,000 tons of polyunsaturated fatty acids (PUFAs) emerging
51 from global streams and rivers to land annually. Adult aquatic insects also transport
52 contaminant loads, including more than 400 tons of metals and tens of kilograms of
53 pesticides each year.

54 Considering that about 90% of insect biomass is deposited within 1 km of rivers ⁷,
55 riparian zones emerge as global hotspots of both nutrient and contaminant transfer
56 by adult aquatic insects. These findings reveal aquatic insects as overlooked vectors
57 of global biogeochemical fluxes, with critical implications for food-web dynamics and
58 ecosystem health.

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66 MAIN

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68 Animal movement and migration redistributes biomass and nutrients within and across
69 ecosystem boundaries, often supplying a substantial portion of the nutrient budget of
70 recipient communities ^{2,8}. Freshwater insects develop as larvae within aquatic ecosystems
71 and subsequently emerge as terrestrial winged adults. The emergence of adult aquatic
72 insects represents a key source of prey biomass and essential nutrients (e.g., nitrogen,
73 phosphorus, and polyunsaturated fatty acids [PUFAs]) to a range of terrestrial predators,
74 including birds, bats and spiders ^{3,5,9}. This export of biomass and nutrients can trigger
75 cascading effects across trophic networks, potentially increasing the breeding success and
76 growth rate of riparian consumers ^{10,11} and influencing overall stability of aquatic-terrestrial
77 food webs ¹². Adult aquatic insects emerge from all major waterbodies including lakes,
78 wetlands and streams and rivers; but rivers and streams tend to have the highest
79 proportional export to land because of their lower surface area to shoreline ratios (Sabo and
80 Hagen 2012). As such, >90% of aquatic insects exported from inland waters are deposited
81 in terrestrial ecosystems, with rates comparable, for instance, to secondary production of
82 herbivorous insects in grasslands ^{13,14}.

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84 In addition to transporting nutrients and biomass from water to land, aquatic insects also
85 export contaminants. Aquatic ecosystems sit topographically lower in the landscape and
86 receive various contaminants from adjacent terrestrial ecosystems through runoff ¹⁵ or
87 effluent discharges ¹⁶. Consequently, reducing the effects of pollution on aquatic ecosystems
88 has been a major focus of contaminant regulation and policy efforts ¹⁷. For example, an
89 estimated US\$300 billion annually is spent on mitigating global water pollution to reduce
90 effects within aquatic ecosystems ¹⁸. However, contaminants in aquatic ecosystems can be
91 also transported back to terrestrial ecosystems by abiotic mechanisms such as floods ¹⁹, or
92 by organisms such as aquatic insects ¹. The associated export of waterborne contaminants
93 bioaccumulated in aquatic insect larvae and retained across metamorphosis in the tissues of
94 adult aquatic insects can move bioavailable contaminants to terrestrial ecosystems, possibly
95 counteracting the benefits of a nutrient-rich prey source to terrestrial consumers ^{3,4}. For
96 instance, more than 6 g yr⁻¹ of PCBs were exported to terrestrial spiders feeding on
97 emerging aquatic insects along 25 km of a contaminated stream ⁴. This export is equivalent
98 to the polychlorinated biphenyls (PCB) mass transported by over 50,000 yearly returning
99 Chinook salmon in the polluted Duwamish River in Washington State (USA) ^{4,20}. Despite the
100 role of aquatic insects in transporting bioaccumulated contaminants from aquatic to

101 terrestrial ecosystems^{1,4}, the magnitude and spatial patterns of adult aquatic insect
102 emergence, and their simultaneous export of contaminant and nutrient, have not been
103 estimated at the global scale.

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105 Because the balance between nutrient subsidies and contaminant exposure likely varies
106 geographically, spatially explicit assessment can reveal locations where nutrient-contaminant
107 ratios may lead to ecological compensation and trade-offs²¹. Ultimately, spatial quantification
108 of the export of emerging insects from aquatic ecosystems is needed to enhance our
109 understanding of the ultimate fate, exposure and effects of subsidies and contaminants
110 through food webs and across ecosystems^{1,22,23}.

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112 Here we estimate global patterns of biomass, nutrient, and contaminant export in adult
113 aquatic insects emerging from streams and rivers. To do so, we use 305 globally distributed
114 records of aquatic insect production, information on nutrient concentrations in adult aquatic
115 insects, and published models of global surface water area²⁴ to calculate aquatic insect
116 emergence biomass and nutrient export from global streams and rivers. Then, using >13
117 million observations from continental scale water quality monitoring stations, we model water
118 column concentrations for selected waterborne contaminants, including metals and
119 metalloids (hereafter together referred to as metals) and pesticides, for >1 million river
120 basins across the globe. Finally, we combined the water contaminant models with
121 empirically-derived transfer functions that link water column contaminant concentrations to
122 concentrations in the tissue of emergent adult insects (534 records) to estimate and map the
123 global export of waterborne contaminants from water to land.

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126 RESULTS AND DISCUSSION

127 *Emergence-mediated biomass and nutrient export*

128 Scaled by the total surface area of streams and rivers on Earth²⁴, our models estimate that
129 the median annual global export of insect biomass (measured as dry mass; DM) from water
130 to land is 1,820,000 t yr⁻¹ (95% Credible Interval (CrI) = 1,232,000–3,160,000, **Tab. 1**). These
131 estimates derive from generalized hierarchical Bayesian modelling that combined empirical
132 observations of aquatic insect annual benthic secondary production and adult aquatic insect
133 emergence biomass in flowing waters (as mg dry mass m⁻² yr⁻¹) covering all major biomes

and continents (**Extended Data Fig 1**), with a range of climatic and geospatial predictors (Methods). Model selection indicated that biomass of adult aquatic insect emergence within watersheds was effectively predicted based on mean annual temperature and annual precipitation (**Extended Data Fig. 2**), in line with earlier continental-scale synthesis literature²⁵. The model predicted the median dry mass of insects leaving streams and rivers to be 1,680 mg m⁻² yr⁻¹ (95% CrI = 130–17,770; **Tab. 1**). Because temperature and precipitation ranges in the training data were roughly comparable to global ranges (**Extended Data Tab. 1**), we extrapolated our model predictions to the nearly 1 million river basins (HydroBasin level 12; hereafter “hydrobasin”; see Methods) in the HydroSheds database²⁶.

Tab. 1 - Modeled estimates of dry mass and nutrient export from water to land by adult aquatic insects in export per square meter and global export per year. Values represent the posterior median and upper and lower 95% CrI. Values are rounded to the nearest 10 for export per area and nearest 1000 for global export. PUFA = polyunsaturated fatty acids.

Nutrient	Units	Median	Q2.5	Q97.5
<i>Export per surface water area</i>				
Dry Mass	mg·m ⁻² ·y ⁻¹	1,680	130	17,770
Carbon	mg·m ⁻² ·y ⁻¹	750	60	8,000
Nitrogen	mg·m ⁻² ·y ⁻¹	120	10	1,270
PUFA	mg·m ⁻² ·y ⁻¹	20	0	210
Phosphorus	mg·m ⁻² ·y ⁻¹	10	0	60
<i>Global export</i>				
Dry Mass	t·y ⁻¹	1,820,000	1,232,000	3,160,000
Carbon	t·y ⁻¹	819,000	555,000	1,422,000
Nitrogen	t·y ⁻¹	130,000	88,000	226,000
PUFA	t·y ⁻¹	17,000	4,000	69,000
Phosphorus	t·y ⁻¹	7,000	4,000	11,000

Other efforts have quantified insect emergence at finer spatial scales and serve to validate our global analysis. For example, our global median estimate of $\sim 1,600 \text{ mgDM m}^{-2} \text{ yr}^{-1}$ of adult aquatic insect production is roughly equivalent to $\sim 750 \text{ mg carbon (C) m}^{-2} \text{ yr}^{-1}$, and is consistent with previous estimates of $\sim 1,000 \text{ mgC m}^{-2} \text{ yr}^{-1}$ by Gratton and Vander Zanden¹³, and within the range estimated by Bartrons *et al.*²⁷ for lotic systems in the state of Wisconsin (< 0.0001 to $9,000 \text{ mgC m}^{-2} \text{ yr}^{-1}$). Similarly, emerging insects are estimated to transport approximately 1,400 tons of carbon from Wisconsin's rivers annually²⁷, which corresponds to about 1/650th of our global estimates and is close to the estimated proportion of river surface area in Wisconsin relative to the entire Earth (1/570th). This provides additional support for our global estimates based on local measurements, and also suggests that the density and surface area of rivers within basins are the primary determinants of total insect emerging biomass. Spatially scaling our estimates of insect production to the total surface area of streams and rivers within each hydrobasin (**Fig. 1**) indicates that, besides tropical regions, export of emerging adult aquatic insects is particularly abundant in small coastal basins, such as in British Columbia, Norway and Chile, and throughout the Arctic. This reflects the relatively high density of flowing water in these basins, as derived from recent global calculations^{24,28}, even though Arctic basins are less productive per unit area of water than other regions (**Fig 2; Extended Data Fig. 3**). Our estimates also highlight the relatively high secondary production (per unit of water surface area) of desert and temperate streams, which can be equal to or higher than tropical streams^{14,29,30}. This non-intuitive pattern is likely due to higher fish predation rates for larger insect larvae, and the relatively higher standing biomass of non-emerging invertebrates (i.e. crustaceans, molluscs) in tropical rivers^{30,31}.

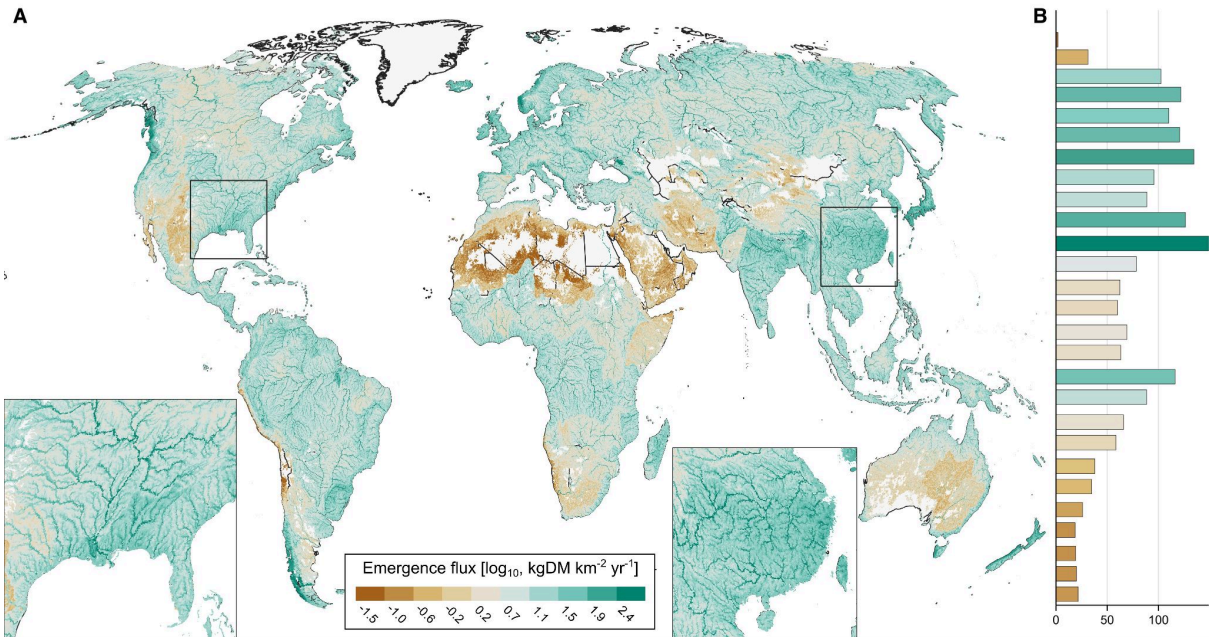
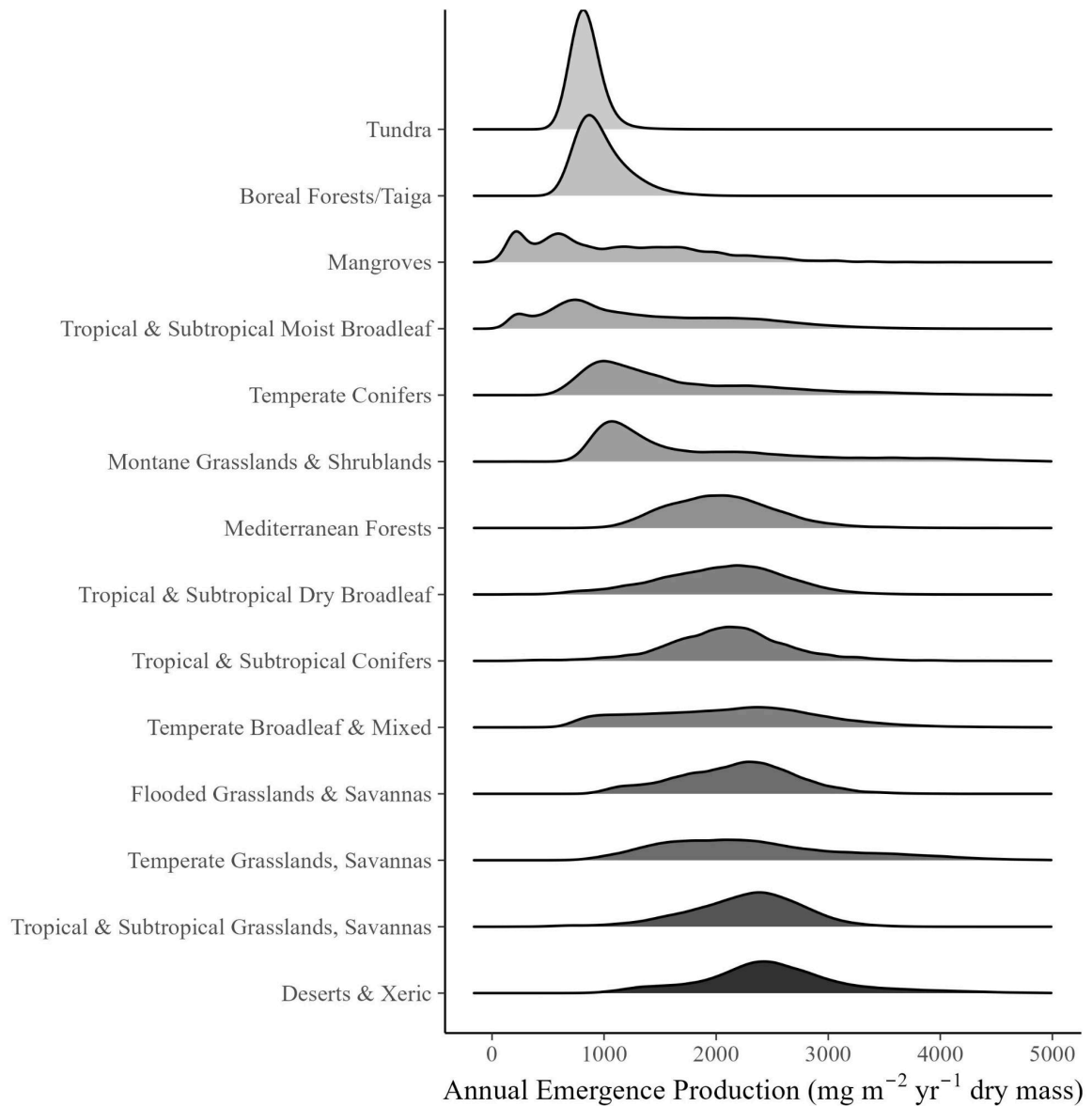


Fig 1 - (A) Global map of insect emergence biomass in $\text{kgDM km}^{-2} \text{ yr}^{-1}$ (basin area) derived per hydrobasin ($n = 919,156$) and extrapolated using hierarchical Bayesian modelling (305 reports, from 111 studies). Zoom-panels depict the Mississippi Portal and Southeast China. Values are displayed on a $\log_{10}$ -scale for ease of interpretation and rasterized in 0.05° resolution. (B) Bargraphs depict the total annual insect emergence biomass in kilotons per 5° longitude bin.



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179 **Fig. 2.** Annual emergence production as mgDM m^{-2} surface area of water yr^{-1} for the Earth's
 180 terrestrial biomes. Densities show the distribution of posterior median emergence production across
 181 hydrobasins in each biome. The number of hydrobasins ranges from 3,046 in Mangroves to 158,638
 182 in Tropical & Subtropical Grasslands, Savannas. Densities are truncated at 5,000 $\text{mgDM m}^{-2} \text{yr}^{-1}$ to
 183 improve visibility. The color of each density corresponds to its median production on a gray scale..

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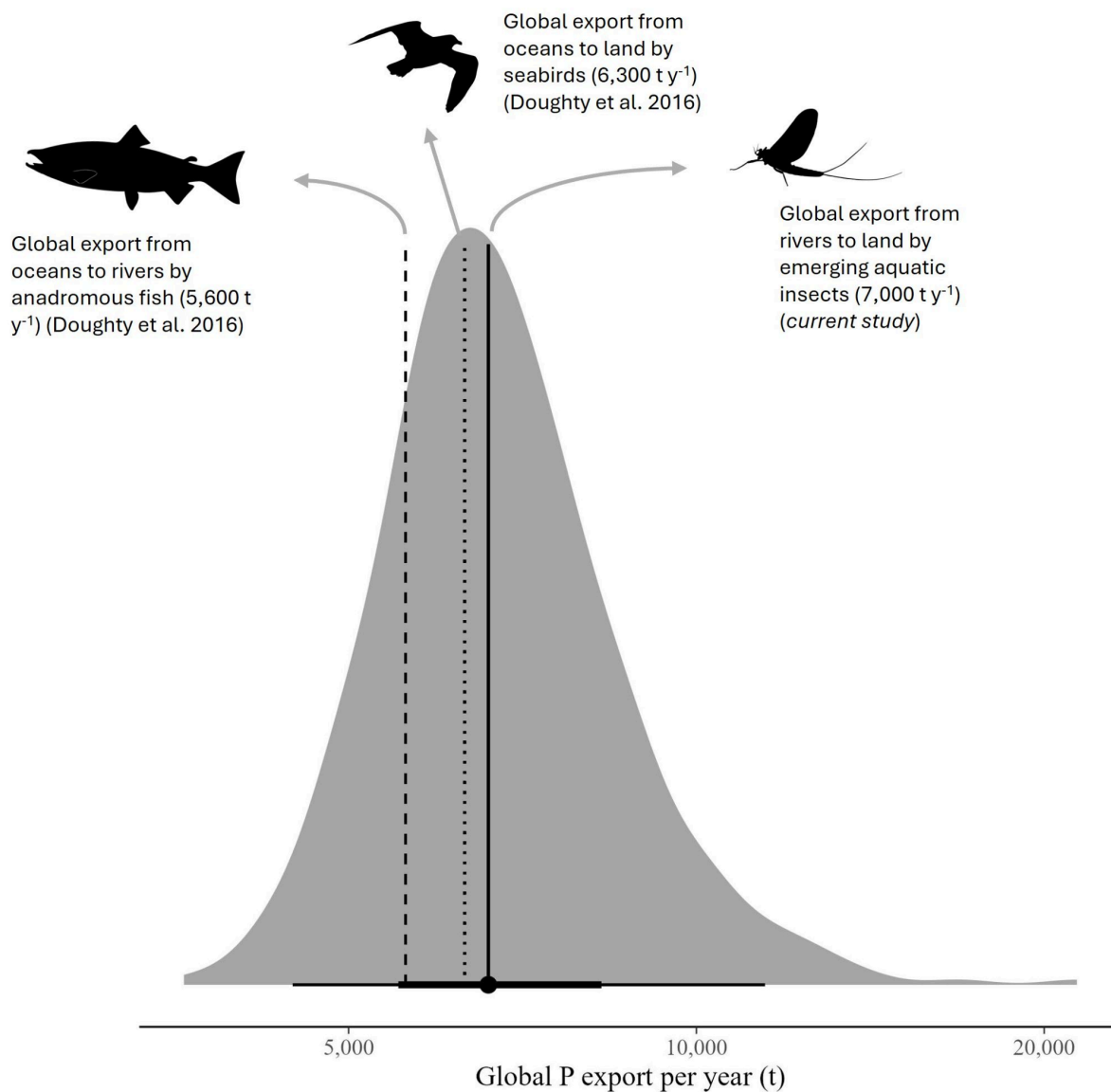
187 Our results indicate that annual aquatic insect emergence production is among the largest
 188 contributors of animal-mediated nutrient flow ever reported (**Fig.3**). The estimated ~1.8
 189 million tonnes of dry mass of emerging insects corresponds to about 0.8 million tonnes of
 190 carbon (**Tab. 1**). As a comparison, terrestrial arthropods make up 0.2 gigatons of carbon, or
 191 about 200 million tonnes ³². Thus the yearly export of adult aquatic insects represents

192 ~0.45% of the standing mass of terrestrial arthropods (i.e., 0.8 million t / 200 million t =
193 0.4%), similar to the percent of river water on Earth relative to non-glaciated land surface
194 (0.59%, Allen and Pavelsky 2018). Using standard C:N (6.3:1) and C:P (124:1) ratios for
195 insects ³³, we estimate that adult aquatic insects export ~130,000 tonnes of N and 7,000
196 tonnes of P from streams and rivers globally (**Tab. 1**). By comparison, approximately 5,600
197 tons of P were estimated to be distributed inland globally by anadromous fish ³⁴. Similarly,
198 recent calculations based on fisheries data indicate that Pacific salmon mobilize about 9,000
199 tonnes of N and 1100 tonnes of P within continental North America ³⁵. Seasonal migrations
200 of large herbivores such as hippopotami and wildebeest are estimated to deliver about
201 30,000 kg of P to the Mara river system in Tanzania ⁸, and desert locusts can mobilize more
202 than 2,350 kg of N and 225 kg of P across northern Africa ³⁶. In all of these regional
203 examples, nutrient export by other animals is exceeded by export from emerging aquatic
204 insects: 121,000 kg of P in western North America, more than 75,000 kg P exported from
205 Tanzania rivers, and about 29,000 kg of P from rivers in Morocco and Algeria. These
206 examples underscore the magnitude and potential importance of aquatic insect emergence
207 to terrestrial ecosystems, even in arid regions where surface water is scant. Moreover, our
208 analyses likely underestimate the overall magnitude of emergent biomass since our
209 estimates do not include extensive and highly productive aquatic habitats such as lakes and
210 wetlands at the global scale.

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216 **Fig. 3** - Posterior distribution of phosphorus (P) export by emerging aquatic insects in freshwater rivers
 217 compared to anadromous fish and seabirds. The gray density represents the posterior distribution of
 218 insect-mediated P export. The dot and horizontal error bars at the bottom show the median with 66%
 219 and 95% credible interval (CrI). The vertical solid line also shows the median P export by emerging
 220 aquatic insects. The dashed and dotted lines show export of P from oceans to rivers by anadromous
 221 salmon or from oceans to land by seabirds via guano deposition, estimated from ³⁴.

222

223 Aquatic food webs are generally richer in long-chain polyunsaturated fatty acids (LC-PUFAs)
 224 compared to terrestrial food webs ³. As a result, emerging insects serve as key sources of
 225 these fats, and are indispensable for secondary production in terrestrial systems. Our
 226 estimates show that between 4,000 and 69,000 (median 17,000) tonnes of PUFAs,
 227 particularly eicosatetraenoic acid (EPA) and docosahexaenoic acid (DHA), are exported
 228 annually from Earth's rivers. These values correspond to median export of about 20 mg m⁻²

229 surface water yr^{-1} , in line with values reported in available site-specific studies ^{37,38}.
230 Considering only hydrobasins draining the Pacific in North America, emerging adult aquatic
231 insects are estimated to export ~360 tons (median) of PUFAs (95% CrI: 73–1,200). By
232 comparison, migrating Pacific salmon are estimated to deliver about 3800 tonnes of PUFAs
233 within continental North America ²¹. The importance of PUFA inputs by anadromous salmon
234 to freshwater ecosystems underscores the global role adult aquatic insects likely play as a
235 large yet understudied vector of essential fatty acids from rivers to land ^{3,9,39}.

236 Considering that most emergent insect biomass is deposited within ~1 km inland from river
237 and stream banks ⁷, riparian zones represent vital feeding areas for many terrestrial
238 consumers that connect aquatic and terrestrial meta-ecosystems globally ^{39–41}. General
239 decline of insect populations has thus also implications for trophic interactions across
240 ecosystem boundaries ^{42,43}. Massive declines and extinctions of large migratory animals
241 have already reduced their capacity to redistribute nutrients to about 6% of their historical
242 capacity, in both terrestrial and marine systems ³⁴. Similar historic estimates of
243 insect-mediated flows are not available and, more generally, trends in aquatic insect
244 abundance appear highly variable ^{44,45}. However, several global change drivers, including
245 warming, artificial illumination and eutrophication are already reducing the production of
246 PUFAs in aquatic ecosystems ^{39,46}. Changes in temperature are also creating nutrient
247 phenological mismatches for riparian predators, whereby peak availability of emerging
248 aquatic prey is increasingly shifted earlier in the season, creating mismatches between
249 nutrient demand and availability ^{47,48}. This highlights how ongoing global change drivers can
250 alter the magnitude, timing and quality of aquatic insect export to terrestrial ecosystems with
251 implications for receiving riparian consumers.

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253 ***Emergence-mediated contaminant export***

254 By far, global export of metals (389 t yr^{-1} , 95% CrI: 120–856) associated with adult aquatic
255 insect emergence exceeds the export of pesticides (**Tab. 2; Fig. 4**). While the mass of
256 pesticides (including insecticides, herbicides and fungicides) exported was much smaller
257 (0.1 t yr^{-1} , 95% CrI: 0.02–0.43), they represent anthropogenically synthesized chemicals
258 designed to have target effects on insects and their food sources (bacteria, fungi, and plants)
259 at low concentrations ⁴⁹. Trace metals and mercury, on the other hand, occur naturally in
260 freshwaters at concentrations ranging from essential or background to toxic for life; although
261 their concentrations can be altered by anthropogenic activities ⁵⁰. These export estimates
262 were obtained by linking projections of contaminant concentrations in global surface waters

with our estimates of global aquatic insect emergence biomass using empirically derived contaminant transfer rates from water to adult emerging insects (see Methods). The modelling of contaminant concentrations in surface water was based on a harmonized dataset that included more than 12.7 million measurements from the United States ($n \approx 5.3$ million) and Europe ($n \approx 7.4$ million) from more than 60,000 locations used for model training, and about 3.2 million measurements from more than 7,700 locations across all continents used for model validation (**Extended Data Fig. 4**). Because harmonized long-term contaminant monitoring datasets are currently most extensive in North America and Europe, model calibration relied primarily on observations from these regions, while datasets from additional continents allowed reducing prediction bias and evaluating model performance at the global scale (see Methods).

There are no other global biota-mediated metal export data available. Total annual metal export to the oceans for Cr, Mn, Co, Ni, Cu, and Zn has been estimated at more than 26 million t yr^{-1} ⁵¹, which is more than two orders of magnitude larger than our global estimate for insect-mediated export to land. Similarly, our estimated insect-mediated pesticide export to land is more than four orders of magnitude smaller than the globally estimated pesticide export from rivers to sea of about 710 t yr^{-1} ⁵², which likely is driven by the approximately 1.8 million times larger volume of terrestrial runoff⁵³ compared with the biomass of emerging aquatic insects. Based on the restricted availability of data, we have considered only the six metals for which data were available in the literature, of the approximately twelve potentially relevant metals and only 25 out of the approximately 1000 potentially relevant pesticides⁵⁴ used globally. Likely, total export estimates will increase when data on insect emergence-mediated transfer rates become available for more compounds.

Most metals exported from streams and rivers by adult aquatic insects are essential metals (total essential metal export = 368 t yr^{-1} , 95% CrI: 110–850; Cu = 54, Se = 5, Zn = 309 t yr^{-1} ; **Extended Data Fig. 5**); non-essential metals comprised about 11 t yr^{-1} (95% CrI: 5 to 49; Cd = 2, Hg = 0.7, Pb = 7.9 t yr^{-1} , **Fig. 4A**). This difference in export magnitude can be explained in part by the fact that essential metals are ubiquitously present in insect tissues while non-essential metals are not. As a result, insects export at least some essential metals from all rivers, and exports of individual essential metals are more similar in their spatial patterns and to biomass emergence patterns than exports of non-essential metals (**Extended Data Fig. 6**).

In models predicting metal export across river basins, emergence biomass explained 90 to 98% of variation depending on the metal analyzed. In the same models, aqueous water concentration explained 0.3 to 8% (**Extended Data Tab. 2**). The relatively small influence of

aqueous water concentrations on metal export is somewhat counterintuitive, but it is consistent with existing knowledge of the mechanisms of metal export in emerging aquatic insects. For example, metals are excreted during metamorphosis in a concentration-dependent manner, reducing variation in adult aquatic insect metal concentrations compared with variation in aqueous concentrations^{55–57}. As a result, basins that vary greatly in aqueous concentrations can have similar adult insect tissue concentrations⁵⁸. In addition, most metals at high enough concentrations reduce emergence, which can lead to higher metal export from lower metal streams^{57,59,60}. Together, these mechanisms predict strong contributions of emergence biomass to metal export at the local scale^{61,62}.

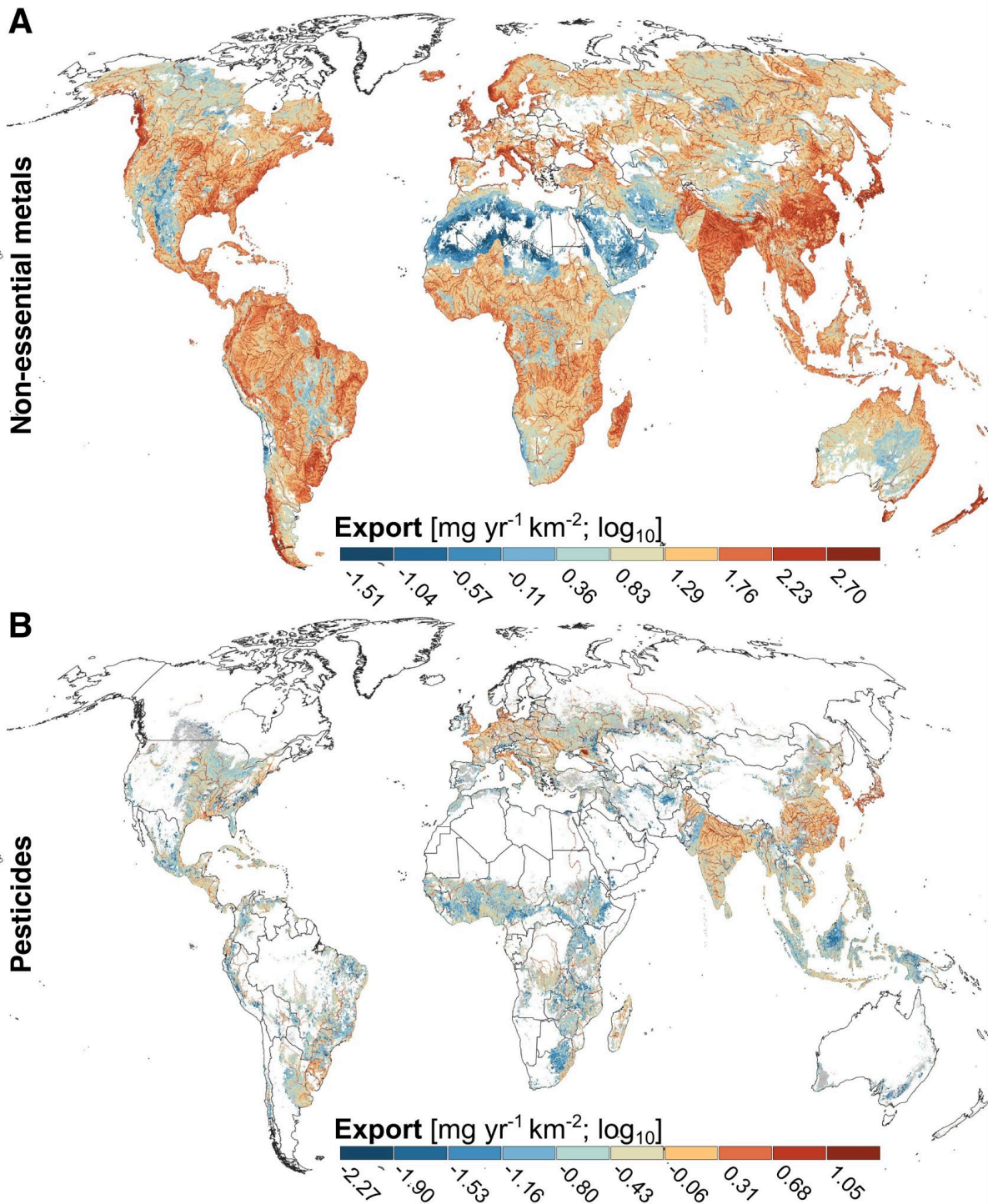
At the spatial scales of our study (i.e., hydrobasins to global), concentration-dependent excretion of metals during metamorphosis rather than mortality effects likely contribute to the observed spatial patterns of contaminant export. The number of basins with toxic concentrations of metals is likely small because our exposure concentrations represent an average across a basin. Within many of those basins, there are almost certainly individual stream reaches with metal concentrations exceeding survival thresholds of insects. At the basin level, however, such streams likely represent a small fraction of total water area and hence almost none of the basins exceed concentrations expected to reduce insect emergence at the basin level. Yet for individual streams where metals reach concentrations that reduce insect emergence, riparian insectivores can suffer from reduced resource availability⁵⁹, rather than increased metal exposure mediated by adult aquatic insect prey^{63,64}.

Although included in our list of non-essential metals, Hg is of special interest because of its global distribution, tendency to biomagnify through the food web, and neurotoxic effects on higher order consumers. The global transport of Hg from river discharge to sea has been estimated at 1000 t yr⁻¹⁶⁵ more than three orders of magnitude higher than our estimated insect-mediated export to land of 0.72 t yr⁻¹. However, we estimated insect-mediated Hg export for the North Pacific US coast to be 21 kg yr⁻¹ (95% CrI: 5–100), three times higher than the 7 kg yr⁻¹ recently estimated for Pacific salmon²¹. Despite being much smaller than physical transport, both salmon and emerging insects represent pathways of great ecological relevance, as they lead to organic Hg being directly available in the riparian recipient food web, which then may biomagnify in terrestrial consumers^{21,66}. The relatively large export of Hg by adult aquatic insects tends to follow spatial patterns of emergence. This finding is consistent with smaller scale studies suggesting that factors like nutrients and fish that

control emergence are strong predictors of Hg export in adult aquatic insects^{62,67}. Together, these results highlight that factors driving emergence determine the global patterns and magnitude of ecologically relevant Hg export from rivers and streams via adult aquatic insects.

Tab. 2 - Modeled estimates of annual global metal and pesticide export from water to land by adult aquatic insects from rivers. Values represent the posterior median and upper and lower 95% CrI. Each row is calculated separately across posterior iterations of a right-skewed posterior; hence sums of the median for each element may not match the totals.

Contaminants	Units	Median	Q2.5	Q97.5
Zn	t·y ⁻¹	309.1	58.8	801.4
Cu	t·y ⁻¹	53.7	34.8	79.6
Pb	t·y ⁻¹	7.9	2.4	46.0
Se	t·y ⁻¹	4.8	1.5	13.7
Cd	t·y ⁻¹	2.1	0.4	6.5
Hg	t·y ⁻¹	0.7	0.2	2.5
Total metal	t·y ⁻¹	389.0	120.6	856.7
Herbicide	t·y ⁻¹	0.086	0.014	0.421
Insecticide	t·y ⁻¹	0.009	0.002	0.037
Fungicide	t·y ⁻¹	0.003	0.001	0.012
Total pesticide	t·y ⁻¹	0.102	0.023	0.438



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343 **Fig. 4** - Global annual aquatic contaminant export in mg km^{-2} (basin area) mediated by aquatic insect
 344 emergence derived from 919,156 hydrobasins. Values are displayed on a $\log_{10}$ -scale for ease of
 345 interpretation and rasterized in 0.05° resolution. Contaminant export was estimated by combining
 346 meta-analytically derived emergence productivity (Bayesian GAMM; $n = 305$ reports from 111
 347 studies) and contaminant transfer (i.e. aqueous uptake) models (GLMM; $n = 534$ reports from 20
 348 studies) with random-forest inferred ($n \sim 8.9$ million, AUC 0.965–0.991) aqueous contaminant
 349 occurrence estimates ($n \sim 8.0$ million stream features, $n = 919,156$ hydrobasins) based on 24

selected spatial predictors and the observation period 2015–2019. For validation and accuracy metrics see Supplementary Methods. Two contaminant groups are included: (A) non-essential metals (cadmium, lead, mercury) and (B) 25 pesticides. Agricultural areas ($\geq 25\%$, 0.05° , grey) are provided in (B) using Corine land-cover estimates for 2018.

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For pesticides, annual exports (**Tab. 2; Fig. 4B**) were lower than metals at approximately 100 kg yr^{-1} (95% CrI: 23–438), and differed by pesticide class, with median areal export rates of 3.5, 115 and 12 mg km^{-2} basin area for fungicides, herbicides and insecticides, respectively. Elevated exports were found predominantly in intensified agricultural regions, such as central Europe, eastern China, northern India and parts of North and South America (southern Brazil), consistent with agriculture acting as the primary source of pesticides into non-target ecosystems^{68,69}. Particularly, low-to-middle income economies have experienced rising pesticide applications in recent decades, following agricultural industrialization⁷⁰, suggesting that aquatic-terrestrial export may increase in the future. In contrast, industrialized agriculture in high-income economies transitioned toward more efficacious pesticides that require less mass of active ingredient per hectare⁴⁹, for example, from organochlorine and organophosphate to neonicotinoid and pyrethroid insecticides⁷¹ that have lower application rates and are thus often underestimated in regular surface water monitoring¹⁷. Dynamically shifting pesticide application patterns will likely affect aquatic to terrestrial pesticide export associated with emerging insects.

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Differences in pesticide export by adult aquatic insects are likely due to differences in their presence in the environment as well as their toxicity and bioaccumulation/magnification within insect-based food webs. Herbicides dominate total global pesticide use (51% versus 22% for insecticides and 22% for fungicides in 2023⁷², but this alone does not explain the higher herbicide export reported here. Likely, the comparably low toxicity of herbicides to aquatic insects contributes to the high export, as even higher body burdens of herbicides are not necessarily associated with mortality in aquatic insects. High toxicity and biomagnification potential can alter export of some classes of current-use fungicides or insecticides by adult aquatic insects. For the neonicotinoid insecticides imidacloprid and thiacloprid, it has been shown that even surface water concentrations in the low ng L^{-1} -range lead to concentrations of 1 to 10 ng g^{-1} dry mass in emerging insects (Diptera, Ephemeroptera, Trichoptera)^{73,74}. When multiplied by our global estimates of aquatic insect emergence biomass, this would translate to about $2\text{--}20 \text{ kg yr}^{-1}$ of estimated neonicotinoid export, which is consistent with our estimated 3 kg yr^{-1} export for insecticides globally. Bioaccumulation of water-phase neonicotinoids in emerging aquatic insects and subsequent biomagnification in predatory terrestrial spiders has been shown to increase contaminant

levels along the aquatic-terrestrial food web by up to three orders of magnitude ⁷⁴. Furthermore, some insecticides, (e.g., pyrethroids) are already toxic at concentrations orders of magnitude lower than many metals ¹⁷, which likely adds to their ecological importance at the aquatic-terrestrial interface despite much lower exports compared to metals.

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Global pesticide export via emerging aquatic insects is likely underestimated in the present study for several reasons: i) only a subset of pesticides was included in our estimates of global surface water contamination; ii) our calculation of surface water contamination was based on measured concentrations, which likely underestimate real exposures, at least in the case of insecticides ⁷⁵; iii) contaminant monitoring data do not well represent the contribution of small, first-order streams, which are widely diffused in catchments and could experience high levels of pesticide pollution ⁷⁶, iv) the monitoring data on which our predictions are based largely come from comparably well-regulated countries, in which environmental contamination is likely less widespread than in less regulated countries; and v), our analyses did not include standing waters, such lakes, reservoirs and wetlands. These standing water systems are both extensive and highly productive, yet their contributions are more spatially concentrated and less diffusely integrated into the terrestrial matrix compared to streams and rivers ²⁷.

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While our synthesis represents the first global estimate of its kind, some limitations remain. First, estimation of global insect secondary production relies on a relatively small number of empirical estimations. However, the 305 records used for synthesis are geographically widespread and cover environmental predictors roughly comparable to global ranges. In addition, some source studies did not specifically distinguish non-emerging insects from production estimates, which may introduce minor bias. Second, while the spatial availability of contaminant data was nearly global, monitoring data were sparse in potentially critical areas such as China. Third, our synthesis was limited to a subset of the many potentially relevant contaminants circulating in the environment. Therefore, our synthesis likely underestimates the actual extent of global contaminants export via emerging insects.

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417

418 **Conclusions**

Our results reveal global patterns in a fundamental trade-off in aquatic-terrestrial linkages: the same emergent insects that subsidize riparian food webs with carbon, nitrogen and essential nutrients such as fatty acids can also act as vectors of metals and pesticides. This dual role implies that the nutritional benefits of insect subsidies may be counterbalanced, or even outweighed, by contaminant exposure depending on local context. In resource-poor

424 riparian systems, emergent insects can provide valuable subsidies of long-chain PUFAs to
425 terrestrial consumers, yet in intensively farmed or industrialized catchments the same
426 biomass export may increase ecotoxicological risks for insectivorous predators.

427

428 These trade-offs are highly heterogeneous across the globe. Relative export maps
429 (**Extended Data Fig. 7**) show that contaminant exports are more spatially concentrated than
430 nutrient subsidies (i.e., PUFAs), because nutrient export scales largely with insect biomass
431 and river surface area, whereas contaminant export additionally depends on aqueous
432 contaminant concentrations that are influenced by catchment-scale land use. This results in
433 basins where emergent insects deliver disproportionately high contaminant loads relative to
434 their nutritional value. The relative export maps highlight these patterns, showing basins that
435 export relatively more contaminants than PUFAs compared to the global export of both
436 compounds. Understanding and quantifying these trade-offs are essential for predicting net
437 ecological outcomes and for guiding management that seek to balance the benefits of
438 aquatic subsidies against the risks of pollutant transfer across ecosystem boundaries.

439

440 This synthesis highlights aquatic insects as under-appreciated yet globally significant
441 potential vectors of matter and energy across ecosystem boundaries. Because
442 approximately 90% of emergent insect biomass is deposited within roughly 1 km of stream
443 and river banks, these exports are spatially concentrated, creating hotspots where terrestrial
444 predators such as birds, bats and spiders are simultaneously subsidized with nutrients and
445 exposed to contaminants transported via emerging aquatic insects. Given the global length
446 of streams and rivers, these areas make up approximately 50% of the total land surface area
447 on Earth (except in deserts or Arctic areas with low or no insect emergence), highlighting
448 that these exports are globally widespread. Taken together, these findings underscore a
449 potential ecological trade-off with direct implications for food-web dynamics and ecosystem
450 health.

451

452 METHODS

453 *Downscaling estimated rivers and stream surface area*

454

455 We obtained estimates for global rivers and stream surface area (RSSA) from Allen and
456 Pavelsky²⁴. These estimates were produced by combining satellite observations of large
457 rivers (>90m) with a statistical approach to account for the surface area of smaller streams.
458 RSSA estimates are provided as grouped major basins^{24,26}, which often encompass multiple
459 hydrobasins catchment polygons (HydroBASINS level-12; resolution used in the present
460 work). RSSA files remain spatially concordant with hydrobasin polygons, allowing for a
461 one-to-many relationship, although this approach does not adequately capture underlying
462 hydrological patterns, such as variations in stream order or regions with dense fluvial
463 networks. To enhance the spatial allocation of RSSA to each hydrobasin, we downscaled
464 RSSA estimates using the distribution of HydroRivers as a weighting factor. For each
465 hydrobasin level-12, we identified the Strahler orders for all HydroRivers flowlines within and
466 associated them with stream width estimates obtained from Downing et al.⁷⁷. The
467 'Downing-based' surface areas were derived leveraging the established relationship
468 between Strahler order (ω) and stream width (w) as:

469
$$w = 0.452 \cdot e^{0.842\omega}$$

470 where w represents the estimated stream width corresponding to Strahler order ω .

471 Individual, "Downing-based" stream surface areas A were then calculated as:

472
$$A = w \cdot L$$

473 where L is the stream length (derived per HydroRiver feature).

474 Stream surface areas were subsequently aggregated per hydrobasin, rescaled between 0-1
475 and used as weighting factors to proportionally distribute RSSA among the underlying
476 hydrobasins. RSSA were downscaled using the Downing-based surface areas, rather than
477 only estimating global surface areas via Strahler orders, to prevent overestimations of global
478 river surface areas and to incorporate robust remote-sensed information from Allen &
479 Pavelski²⁴. Briefly, although the relationship between Strahler orders and average stream
480 widths is robust, the underlying variability is nonetheless considerable. Therefore, global
481 stream surface areas, which directly define global emergence export, may be overestimated
482 by calculations simply based on Strahler order. Additionally, lower order streams constitute
483 a considerable fraction of the global stream surface areas⁷⁷ but are likely underrepresented
484 in HydroRivers due to the resolution of globally available digital elevation maps, further
485 supporting the use of the validated RSSA data as the primary determinant for our export
486 estimates.

487

488 The available RSSA dataset²⁴ had minor data gaps, predominantly along shorelines,
489 resulting in a globally derived RSSA of ~755,000 km², lower than the ~773,000 km² reported
490 by Allen and Pavelski²⁴. To address these gaps, a correction algorithm was developed in
491 which missing hydrobasins were assigned RSSA values based on the mean percentage of
492 stream area (mn_RSSApc) from surrounding catchments with available data. The RSSA for

missing catchments was then estimated by multiplying mn_RSSApc by the respective catchment area. This correction resulted in a global RSSA of 772,884 km², closely aligning with the ~773,000 km² reported by Allen and Pavelski²⁴.

Insect secondary production and emergence data collection

We collated a global database on secondary production to inform our model of emergence from streams and rivers (<https://github.com/Aqua-Sync/secondary-production>). We began by extracting annual estimates of in-stream, benthic invertebrate secondary production from a recent synthesis²⁵. Due to differences in objectives (e.g., annual vs. seasonal production, production across habitat types), not all data from that synthesis were usable for our study. Further, we added new datasets published since 2020 that were not available for inclusion in that former synthesis; these new datasets and others fitting our objectives were identified in part from Google Scholar and Web of Science searches with the same keywords used by²⁵. We also opportunistically added studies and data from papers and datasets cited within Patrick et al., the new studies, and other tables or values contained therein, as well as from studies identified separately when adding complementary variables from the emergence syntheses discussed below. Data were collated, per site, on total aquatic invertebrate annual secondary production and insect secondary production when available. Data were converted to mg dry mass m⁻² y⁻¹ using published conversion factors from ash-free dry mass⁷⁸, wet mass, or carbon⁷⁹. Papers that reported production for only a subset of aquatic insect taxa were excluded to avoid underestimating total community emergence production.

Aquatic insect production provided the most direct data source for modelling emergence export, as non-insect components of benthic secondary production (e.g., mussels, worms) lack a terrestrial life stage. However, in many cases total invertebrate secondary production data were available but aquatic insect secondary production was not; those total data were used as a proxy for aquatic insect secondary production using a conversion factor (see **Annual insect emergence production**). Our approach likely did result in some non-emergent aquatic insect taxa (e.g., backswimmers, diving beetles), which were impossible to exclude from many papers that provided only a single aquatic insect secondary production estimate, being included as aquatic insect secondary production. Our goal was to provide a robust dataset for parameterizing the annual emergence export model. We excluded datasets that only provided instantaneous, daily secondary production, or annual secondary production estimates that were altered from a reference baseline (e.g. studies that experimentally increased nutrient availability). Our efforts resulted in a dataset of 305 annual secondary production estimates, from 111 studies spanning on every continent except Antarctica, expanding on the data available from Patrick et al.²⁵ by ~50% (**Extended Data Fig. 1**).

Out of the 305 resulting emergence estimates, 293 were derived by converting benthic secondary production to adult aquatic insect annual emergence whereas 18 directly estimated adult aquatic insect annual emergence. Our conversions from benthic production produced reasonable estimates of adult emergence. As shown in **Extended Data Fig. 8**, emergence values that were directly measured are nested within the estimates from benthic secondary production. This suggests that the conversions produced reasonable estimates of annual insect emergence, as observed elsewhere¹³.

537 **Biomass export of aquatic insects**

538 The goal for biomass modelling was to generate predictions of annual insect emergence dry
539 mass (DM) from rivers in each of 919,156 hydrobasins. Note that all processing workflows
540 were initially conducted for all level-12 hydrobasins ($n = 1,034,083$) but subsequently
541 reduced to non-desert, permanent lotic systems ($n = 919,156$, for details see **Global**
542 **modelling of aqueous contaminant concentration**). Annual global emergence was then
543 estimated by summing emergence across hydrobasins. To do this, we first converted
544 literature-derived measures of annual benthic macroinvertebrate secondary production (P_b)
545 to annual insect emergence production (E_i) in units of $mgDM\ m^{-2}\ y^{-1}$.

546 We then estimated non-linear relationships of E_i with environmental predictors and used the
547 parameters from those models to predict $E_{i,basin}$ at each hydrobasin. Finally, we multiplied
548 the predicted $E_{i,basin}$ by the surface area of river water to yield the total emergence

549 production per basin. Finally, summing across all basins $\sum_{basin=1}^{919,156} E_{i,basin}$ yielded the total

550 annual emergence biomass globally in units of kilograms or tons per year. Below we
551 describe each of these steps in sequence. We have followed the notation of Gratton and
552 Vander Zanden¹³. A table of parameter definitions is in **Extended Data Tab. 3**.

553

554

555 **Annual insect emergence production**

556 To convert benthic secondary production to insect emergence, we first calculated the raw
557 fraction of community production composed of insects (f_i) from the 111 studies that reported
558 both community-level and insect-only production. To estimate uncertainty in those
559 proportions, we fit an intercept only model with f_i as the response variable as follows:

560
$$f_i \sim \text{Beta}(\mu, \phi)$$

561
$$\text{logit}(\mu_{f_i}) \sim \text{Normal}(0.66, 0.07)$$

562
$$\phi \sim \text{Gamma}(0.01, 0.01)$$

563 The likelihood was Beta and the prior for the intercept was the $N(0.66, 0.07)$, corresponding
564 to a prior mean fraction of ~66%, consistent with prior estimates by Gratton and Vander
565 Zanden¹³. This produced a posterior distribution of the average fraction of secondary
566 production that is insects μ_{f_i} . We took 1000 k draws from that distribution and multiplied each
567 draw by a corresponding benthic secondary production estimate P_b . The result yields a
568 distribution of aquatic insect-only benthic secondary production P_i^k , such that $P_i^k = P_b \cdot \mu_{f_i}^k$.
569 Averaging over the k draws yielded a mean value of aquatic insect benthic production P_i for
570 each corresponding measure of total secondary production.

To convert P_i to emergence production E_i , we first estimated the posterior distribution of emergence-to-production ratios E:P from streams using data from Gratton and Vander Zanden ¹³ and Raitif *et al.* ⁸⁰, which included multiple studies and taxa. We used a Beta likelihood and an intercept only model with varying intercepts for study and taxon:

$$\begin{aligned}
 E: P &\sim \text{Beta}(\mu, \phi) \\
 \text{logit}(\mu_{E:P}) &= \alpha + \alpha_{\text{study}} + \alpha_{\text{taxon}} \\
 \alpha &\sim \text{Normal}(0.19, 0.05) \\
 \alpha_{\text{study}} &\sim \text{Normal}(0, \sigma_{\text{study}}) \\
 \alpha_{\text{taxon}} &\sim \text{Normal}(0, \sigma_{\text{taxon}}) \\
 \sigma_{\text{study}}, \sigma_{\text{taxon}} &\sim \text{Exponential}(2) \\
 \phi &\sim \text{Gamma}(0.01, 0.01)
 \end{aligned}$$

After obtaining this estimate, we Monte Carlo multiplied 1000 k draws from the posterior distribution of E:P ratios by 1000 k draws from the posterior distribution of insect production estimates $E_i^k = \mu_{E:P}^k \cdot P_i$. Averaging over the draws yielded 293 modeled estimates of median E_i . We combined these with 18 empirical estimates resulting in 305 total estimates of annual insect emergence production E_i ($\text{mg} \cdot \text{DM} \cdot \text{m}^{-2} \cdot \text{y}^{-1}$). For simplicity, we consider that all emerging insects contribute biomass export to terrestrial systems, although a small fraction may be deposited back into the aquatic habitat ¹⁴.

589

Relationships of emergence and environmental variables

To determine which environmental variables best predicted emergence E_i , we used generalized additive models (GAM) with median emergence estimates from the above procedures ($n = 305$) as the response variable and a set of environmental variables as predictors (**Extended Data Tab. 4**). The likelihood was Gamma with a log-link. Prior values were determined with prior predictive simulation ⁸¹. We fit 15 models (**Extended Data Tab. 5**) with predictors that included combinations of mean annual water temperature, mean annual precipitation, and 8 other geospatial predictors. All models had varying intercepts of study and hydrobasin, both of which were included to allow posterior prediction to new locations. Values for the fixed predictors were determined using the Lat/Long location of each emergence sample and the corresponding environmental variables available through HydroAtlas (hydrobasin level-12). The general model structure was:

$$\begin{aligned}
 \frac{E_i}{\max(E_i)} &\sim \text{Gamma}(\mu, \phi) \\
 \log(\mu) &= \alpha + f(X) + \alpha_{\text{study}} + \alpha_{\text{basin}} \\
 \alpha &\sim \text{Normal}(-5, 2)
 \end{aligned}$$

604

$$\begin{aligned}
f(X) &= \sum_{k=1}^K \beta_k b_k(X) \\
\beta_k &\sim \text{Normal}(0, \sigma_f) \\
b_k &\sim \text{Normal}(0, 2) \\
\sigma_f &\sim \text{StudentT}(3, 0, 2.5) \\
\alpha_{study} &\sim \text{Normal}(0, \sigma_{study}) \\
\alpha_{basin} &\sim \text{Normal}(0, \sigma_{basin}) \\
\sigma_{study}, \sigma_{basin} &\sim \text{Exponential}(4) \\
\phi &\sim \text{Exponential}(0.01),
\end{aligned}$$

where the response variable E_i is annual insect emergence production. To improve model fit and simplify the prior selection, E_i was standardized to the 0-1 scale by dividing each value by the maximum E_i prior to fitting. The likelihood is Gamma with an unknown mean μ and shape ϕ . The mean μ is described by an intercept α and varying intercepts for study α_{study} and hydrobasin α_{basin} , along with the thin-plate spline smoothing function $f(X)$ for X predictors. The smoothing function is described by k basis functions b_k and their corresponding weights β_k . The total k varies by model as determined using the algorithms in mgcv (Wood 2025). The remaining lines denote prior distributions for each parameter ($\alpha, \beta_k, b_k, \alpha_{study}, \alpha_{basin}, \phi$) and hyperparameter ($\sigma_f, \sigma_{study}, \sigma_{basin}$).

Model comparison and global predictions of emergence

We compared the 15 models using leave-one-out-cross-validation⁸² with the *loo()* function in brms⁸³. Four of the top five models had mean annual water temperature and/or mean annual precipitation in univariate, additive, or interactive forms. The remaining model was an intercept only model with no fixed predictors. For this reason, we used the model with main effects and interactions of mean annual water temperature and mean annual precipitation to predict E_i in each of the 919,156 hydrobasins in units of $kg \cdot DM \cdot m^{-2} \cdot y^{-1}$. Multiplying this by the surface area of river water in each hydrobasins yielded the total export per basin ($E_{i,basin}; kg \cdot DM \cdot basin^{-1} \cdot y^{-1}$). We did this across 1000 k draws of the posterior distribution. Summing over basins for each draw yielded a posterior distribution of global insect emergence $E_{i,global}$ in units of $tonnes \cdot DM \cdot y^{-1}$.

Range of training data versus global data

Across the samples with emergence estimates, mean annual stream temperature ranged from 1 to 25°C (Extended Data Tab.1). That covered ~96% of the global range of temperatures across all 919,156 hydrobasins. Mean annual precipitation in the emergence

estimates covered a smaller percentage (~40%) of the global range. However, the median and 95% confidence intervals for precipitation (and temperature) had strong overlap with similar bounds (177 to 2,792 mm m⁻² y⁻¹ for the raw data vs 15 to 2,791 mm m⁻² y⁻¹ across the hydrobasins) (**Extended Data Tab.1**). We also compared the distributions of contaminants across basins relative to the basins in which emergence was collected (**Extended Data Fig. 9**) to determine if the emergence samples were taken more often in basins with low aqueous contaminant concentrations. Results showed strong overlap between the aqueous concentrations of contaminants in basins with emergence compared to the global distributions of contaminants (**Extended Data Fig. 9**). Hence, we assume that our emergence estimates (directly measured or derived from secondary production) came from basins that reflected typical environmental conditions as of 2025.

Nutrient export by adult aquatic insects

In addition to dry mass, we estimated exports of carbon (C), nitrogen (N), phosphorus (P), and polyunsaturated fatty acids (PUFAs) by converting dry mass based on conversion ratios. For C, we first converted dry mass (DM) to ash-free dry mass (AFDM): $AFDM = 0.9 \times DM$ ⁷⁹. C was then estimated as $C = 0.5 \times AFDM$ ⁷⁸. N and P were estimated using stoichiometric ratios of 6.3 C:N and 124 C:P⁸⁴.

No standard conversions exist for PUFA. Therefore, we conducted literature searches in ISI Web of Science using search terms including “PUFA”, *Poly-unsaturated fatty acid*”, “aquatic insect*”, and “emergence” to identify articles on the PUFA content of adult aquatic insects. We focused on the years 2019 to 2024, with information before 2019 taken from recent review papers^{9,85}. We identified those articles that reported quantitative PUFA concentrations per unit of biomass of emerging insects. Additional relevant papers were found examining references within the papers directly. We estimated exports of long-chain poly-unsaturated fatty acids (LC-PUFA) by first extracting 32 records of LC-PUFA concentrations from the literature. Of these 32 records, we limited our analysis to 25 that reported summed concentrations of docosahexaenoic acid (DHA) and eicosatetraenoic acid (EPA) in adult aquatic insects. One record was reported as DHA + EPA in “adult aquatic insects”, but all other were reported as DHA + EPA in individual insect Orders, including Diptera (e.g., Chironomidae, Culicidae), Ephemeroptera, Odonata, Trichoptera, and Plecoptera. We used these data to estimate the average concentration of EPA + DHA in adult aquatic insects using the following model:

$$\begin{aligned}
 F_s &\sim \text{Gamma}(\mu, \phi) \\
 \log(\mu_{F_s}) &= \alpha + \alpha_{study} + \alpha_{taxon} \\
 \alpha &\sim \text{Normal}(-3, 1) \\
 \alpha_{study} &\sim \text{Normal}(0, \sigma_{study}) \\
 \alpha_{taxon} &\sim \text{Normal}(0, \sigma_{taxon}) \\
 \sigma_{study}, \sigma_{taxon} &\sim \text{Exponential}(2) \\
 \phi &\sim \text{Gamma}(0.01, 0.01),
 \end{aligned}$$

677 where F_s is the standardized (z-score) adult tissue concentration of DHA + EPA
678 (standardized by dividing by the maximum concentration), with varying intercepts for study
679 α_{study} and taxon α_{taxon} . Because tissue concentrations are positive and continuous, we used
680 a gamma likelihood with a log-link. Given the standardized data and log-link, we used a $N(-3,$
681 $1)$ prior for the intercept and $N(0, 1)$ prior for the slope, with an exponential(2) prior for the
682 hyperparameter σ_{study} and $\text{Gamma}(0.01, 0.01)$ for ϕ .

683 After un-standardizing, this yielded a posterior distribution of PUFA concentrations in units of
684 $ng \cdot mgDM^{-1}$. To convert this to annual export, we multiplied each estimate of dry mass
685 export by $k = 100$ draws of PUFA, then took the median of across the 100 draws, yielding
686 estimates of PUFA export in units of mg or $kg \cdot y^{-1}$.

687

688 **Global modelling of aqueous contaminant concentration**

689

690 *Monitoring Data* - To estimate global export of contaminants by adult aquatic insects, we first
691 assembled a global dataset on the occurrence and concentration of waterborne
692 contaminants. Monitoring data detailed the occurrence of 37 contaminants (12 metals, 7
693 fungicides, 8 herbicides, 10 insecticides) in aquatic ecosystems, compiled for European
694 countries and the United States. Both essential and non-essential metals were considered
695 contaminants in our analyses, although at low concentrations, essential metals could be
696 considered as a subsidy. European monitoring data⁸⁶ were generated as part of the Water
697 Framework Directive⁸⁷, active since approximately 2000, and processed according to the
698 methodology described in⁶⁹. This dataset comprised 35,781,777 measurements in total,
699 which was reduced to 7,417,367 measurements of the 37 contaminants. For the United
700 States, data were accessed via the Water Quality Portal⁸⁸ and processed following the
701 approach outlined in¹⁷, resulting in 5,338,555 contaminant measurements. The European
702 and US datasets were harmonized (e.g. using standardized CAS-RN attribution, detection
703 classification) and merged resulting in the combined dataset hereafter referred to as
704 “monitoring data”. Further filtering excluded records predating 2010 to minimize legacy
705 contamination impacts on predictions and limited data to lotic systems. The final harmonized
706 dataset encompassed 8,933,938 monitoring records in total, taken from 48,205 locations
707 between 2010 to 2023.

708 *Spatial attributes* - Globally available spatial attributes relevant to the occurrence of
709 contaminants in surface waters – spanning hydrological, litho-geological,
710 land-use/land-cover, and meteorological characteristics – were reprojected to EPSG:4326
711 (WGS84) in their native resolution and assigned to monitoring sites using 5 km buffer zones
712 (**Extended Data Tab. 6**). This approach demonstrated good predictive power while
713 minimizing computational demands compared to hydrological network-based methods (data
714 not shown). Land-cover estimates were aggregated into six distinct classes: cropland, closed
715 forests, open forests, shrubs and grasses, urban areas, and water surfaces^{69,89}. Additionally,
716 each site was assigned its predominant lithological class to limit predictor complexity. Efforts
717 to reduce predictor complexity and inter-correlation were prioritized, as these can hinder the
718 accuracy and generalization of random forest models^{90,91}.

719 Additional spatial attributes from HydroSHEDS datasets were attributed to monitoring sites
720 at the catchment or sub-catchment scale. An initial screening of 55 potentially relevant

HydroATLAS attributes (e.g., discharge, runoff, elevation, air temperature, soil erosion, and population density) identified those with high explanatory power while eliminating those with high inter-correlation or low predictive relevance. Ultimately, nine HydroSHEDS attributes (at hydrobasin level-12) were retained for model building: upstream catchment area, annual average discharge, Strahler stream order, annual air temperature, annual precipitation, terrain slope, soil erosion, land-surface runoff, and population density. To account for the influence of mining activities, two global mining footprint layers were harmonized into a single predictor representing the average mining footprint (**Extended Data Tab. 6**). In total, 24 spatial predictors were included in the final model training.

Model training and tuning - For the global modelling, contaminants were grouped into four categories (herbicides, fungicides, insecticides, and metals). For each group, two random-forest models⁹² were developed: an occurrence model to predict the likelihood of contaminant presence and a concentration model to estimate concentrations where presence was predicted. Random-forest models were chosen based on their successful application in both water and air pollution modelling^{93–95}. Model training and tuning followed three key steps: predictor reduction, hyperparameter tuning, and tree reduction, utilizing *k*-fold cross-validation throughout (*k* = 3). To manage computational demands, stratified subsets (*n* ≤ 500,000; stratification by detection frequency and chemical identity) were used for model tuning, with data split into training (70%) and hold-out testing (30%) datasets. Feature elimination, hyperparameter tuning and tree reduction were applied successively as opposed to brute-forced randomized grid searches applied for substantially smaller datasets at the global scale^{94,96}. Alongside spatial predictors, models incorporated chemical identity (CAS-RN), year, and month of observation to account for substance-specific, annual, and seasonal variability.

Predictor reduction - Recursive feature elimination (RFE) was used to iteratively remove the least important predictors, as assessed by metrics such as mean explained variance or GINI permutation importance. Resulting models were compared to the initial full model (i.e., the model including all predictors), with optimal predictors selected based on out-of-bag error (OOB; occurrence models) or mean absolute error (MAE; concentration models). While error rates on *k*-fold testing data were also evaluated, they were not used for model selection to avoid data leakage. The finalized predictor sets were carried forward to subsequent tuning steps.

Hyperparameter tuning - To reduce computational demands, the number of trees was minimized. Models were trained with varying tree counts (50–400), and the simplest model achieving predictive performance within 1% of the most accurate iteration was selected. This approach ensured negligible accuracy loss while improving model efficiency. Four subsequent hyperparameters were considered for optimization: number of variables per split, minimal node size, maximum tree depth, and sampling fraction using OOB errors and MAE as evaluation metrics depending on the model type. Contaminant data are highly heterogeneous in space, time and chemical reliability^{17,69}. As such, overly optimized models (e.g. via minimum node size, maximum tree depths) can be biased toward the underlying data structure, thereby failing to generalize contamination patterns. Hence, hyperparameters were tuned to improve prediction accuracies while simultaneously maximizing generalization performance (e.g. via max tree reduction) accepting minimal accuracy trade-offs (i.e. 1% relative accuracy loss per hyperparameter compared to the best model).

766 *Final model building and validation* - Using optimized predictors and tuned hyperparameters,
767 final occurrence ($n = 4$) and concentration ($n = 4$) models were trained on the full training
768 dataset (70% of the total data). Model performance was evaluated using both internal
769 metrics – such as permuted OOB errors and MSE from training – and external metrics from
770 the hold-out dataset (30%). Testing accuracies were either comparable to or slightly better
771 than training metrics, indicating no evidence of overfitting and suggesting robust
772 generalization. A comprehensive summary of model accuracy metrics is presented in
773 **Extended Data Tab. 7** and **Extended Data Tab. 8**. Receiver-operator curves (AUC
774 0.965–0.991) for the contaminant occurrence models also suggested high classification
775 accuracy and are shown in **Extended Data Fig. 10**.

776 *Global model application* - The contaminant occurrence and concentration models were
777 applied to the global stream network, represented by the HydroRIVERS and HydroBASINS
778 datasets. Flowlines outside continental boundaries, such as near-coastal estuarine systems,
779 and those within lentic systems identified via HydroLAKES, were excluded from the
780 predictions, resulting in 8,030,568 stream features. For modelling purposes, a theoretical
781 monitoring site was assigned to each stream feature, providing an even representation of
782 global lotic systems. Geospatial predictors for these theoretical sites were derived following
783 the same procedures used for the training data (see above).

784 Monthly contaminant predictions were conducted for the period 2015–2019 to account for
785 temporal variability, resulting in ~17 billion occurrence and subsequent concentration
786 predictions. Detection frequencies and average concentrations were calculated per time-step
787 (e.g. June 2015) and hydrobasin (level-12, the highest spatial resolution, $n = 919,156$) and
788 subsequently averaged over time, yielding average contaminant concentrations (2015–2019)
789 and their likelihood of occurrence (predicted detection frequency). These hydrobasin-level
790 averages of contaminant concentrations and detection frequencies formed the basis for the
791 analyses of contaminant transfers (see Methods section **Contaminant transfer to adult**
792 **aquatic insects**). Our approach using average contaminant concentrations, thus represents
793 a conservative approach of estimating general contaminant patterns, which do not reflect
794 localized contamination peaks potentially found in individual streams. Furthermore, while our
795 training data represents a wide range of bioclimatic regions, model applications can still
796 result in implausible results, as observed for arid or desert regions and highly remote areas
797 in the tropics.

798 The occurrence of pesticides in remote areas was particularly associated with higher
799 uncertainties (**Extended Data Fig. 11**; see below for further uncertainty discussions). Thus,
800 pesticide predictions were restricted to areas with at least residual agricultural influence
801 (agricultural land-use > 5%) either locally (i.e. per hydrobasin) or regionally (based on the
802 entire upstream area) to prevent overestimating pesticide exports. The latter criteria was
803 applied as previous work has shown that natural or protected areas without noteworthy local
804 agricultural land-use can still be exposed through fluvial inflows from upstream agricultural
805 areas see 89 for details.

806 The training data mainly describe contaminant occurrences in permanent lotic systems, with
807 the assumption that water bodies are present. Hence, contaminant predictions were
808 restricted to hydrobasins with an estimated average discharge of $>0.1 \text{ m}^3 \text{ s}^{-1}$. While
809 ephemeral, semi-permanent or transient streams can already occur in ranges of
810 $\sim 0.005\text{--}0.02 \text{ m}^3 \text{ s}^{-1}$ ^{97–99}, these criteria were not conservative enough to identify desert

environments via HydroATLAS attributes based on associated discharge estimates¹⁰⁰. Thus, a stricter cutoff criterion of $0.1 \text{ m}^3 \text{ s}^{-1}$ was used to prevent overestimating primarily metal concentrations. Furthermore, hydrobasins permanently covered by ice or snow (derived via HydroATLAS and Corine land-cover) and those experiencing <2 months of non-freezing temperatures were removed as well. Aforementioned filtering criteria were also applied to all productivity export estimates (e.g. N-export, mgDM-export), resulting in 919,156 viable hydrobasins overall. Resulting contaminant predictions are given in **Extended Data Fig. 12**, shown as long-term exposures. Long-term exposures represent average predicted concentrations adjusted by their likelihood of occurrence (detection frequency); as such, transient but high contaminant concentrations are proportionally downweighted to account for their transience.

Global model validation - Model predictions may be less accurate in regions with distinct environmental or regulatory contexts (e.g., Africa or South America), where homogeneous monitoring data are sparse or completely unavailable. The issue of limited generalization performance when leaving parameter rooms of the training data has been raised¹⁰¹. The GEMStat database, curated by the German Federal Institute for Hydrology, was utilized as an independent validation dataset to address this limitation. GEMStat compiles global surface water monitoring data from various national programs and was excluded from all training workflows. This exclusion reflects the limited availability and heterogeneity of monitoring data outside Europe and the US, particularly in developing regions, and ensures GEMStat serves as an independent dataset for global validation. The dataset, harmonized and error-checked, was reduced to the 37 contaminants modeled in this study, yielding ~3.2 million measurements. Of these, ~1.5 million measurements, spanning all continents, were from outside the training regions. Among the 37 contaminants, 32 were present in the GEMStat data, with the majority of measurements (87.7%, ~2.8 million) related to metals. Outside the training area, pesticide occurrence data were available for 14 substances (mainly from Canada, China, Japan and Uruguay), with just two (chlordane, metolachlor) having more than 100 measurements. Consequently, validation metrics predominantly reflect the generalization accuracy of the metal contaminant models. Nevertheless, pesticide models demonstrated performance on par with the metal models (see **Extended Data Tab. 7** and **Extended Data Tab. 8**), suggesting that their generalization accuracy may also be robust across global contexts.

Predicted average concentrations per hydrobasin were compared against GEMStat data both on actual (i.e. daily measurements) and average concentrations. The latter did not show noteworthy differences and hence will not be discussed further. All comparisons with GEMStat data were made for 2015 onwards to provide a temporally similar basis. Overall, concentration predictions were highly accurate within the training area, as 91.7% of predicted concentrations were within one order of magnitude of the actual observed concentrations (see **Extended Data Fig. 13**). Concentration predictions outside the training area were less but still comparatively accurate as 77.3% of predicted concentrations were within one order of magnitude of the actual concentrations. Residuals were virtually zero-centered on average, with no systemic under- or overestimations, although with slight tendencies toward an underestimation of contaminant concentrations. Taken together, validation with the independent GEMStat data suggested plausible results of average concentrations globally. Finally, we quantified predictor uncertainties spatially to highlight areas with assumed lower reliability (**Extended Data Fig. 11**). Briefly, all spatial predictors were normalized (z-score transformation) with respect to the training data and subsequently

aggregated (mean absolute difference) per hydrobasin. Uncertainty is thereby expressed as the mean absolute standard deviation per area. Also, predictor distributions for both the training and global prediction data are provided in **Extended Data Fig. 14** to depict the representativeness of individual predictors used for global contaminant modelling.

Contaminant transfer to adult aquatic insects

The goal of contaminant transfer modelling was to convert dry mass estimates from emergence biomass modelling into tissue concentrations of metals, herbicides, fungicides, and pesticides. To do this, we first extracted paired water concentrations and adult tissue concentrations of contaminants from 17 studies in the literature, resulting in 524 observations of 9 contaminants (**Extended Data Fig. 15**). We then modeled this relationship using generalized linear mixed models (separate model for each contaminant), yielding predictions of adult tissue concentrations C_a in units of $ng \cdot mgDM^{-1}$ as a continuous function of water concentrations C_w . In this analysis we do not consider the fraction of contaminants that could be offloaded by adult insects via oviposition.

To scale this globally, we generated posterior predictions of C_a in each of the 919,156 hydrobasin using the mean water concentration (C_w) in each HydroBasin derived from contaminant modelling (see **Global modelling of aqueous contaminant concentration**). This generated basin level predictions of adult tissue concentrations of contaminants $C_{a,basin}$. Multiplying these by estimates of adult aquatic insect emergence $E_{i,basin}$ yielded predictions of contaminant export in each hydrobasin $C_{f,basin}$, which we report in units of mg or $kg \cdot y^{-1}$.

To estimate the relationship between water concentrations C_w and adult concentrations C_a of metals, herbicides, fungicides, and insecticides, we fit separate generalized linear mixed models for each compound with the following structure:

$$\begin{aligned} \frac{C_a}{\max(C_a)} &\sim \text{Gamma}(\mu, \phi) \\ \log(\mu) &= \alpha + \beta C_w + \alpha_{study} \\ \alpha &\sim \text{Normal}(-3, 1) \\ \beta &\sim \text{Normal}(0, 1) \\ \alpha_{study} &\sim \text{Normal}(0, \sigma_{study}) \\ \sigma_{study} &\sim \text{HalfStudentT}(3, 0, 2.6) \\ \phi &\sim \text{Gamma}(0.01, 0.01), \end{aligned}$$

where C_a is the adult tissue concentration of a contaminant (standardized by dividing by the maximum concentration), C_w is the water concentration of the contaminant (standardized as a z-score), with varying intercepts for study. Because tissue concentrations are positive and continuous, we used a gamma likelihood with a log-link. Given the standardized data and log-link, we used a $N(-3, 1)$ prior for the intercept and $N(0, 1)$ prior for the slope, with a default priors of $\text{HalfStudentT}(3, 0, 2.6)$ for the hyperparameter σ_{study} and $\text{Gamma}(0.01, 0.01)$ for ϕ .

895 **Model fitting and model checking**

896 Analysis of regression models (simple to generalized linear mixed models) was conducted in
897 R version 4.5.2 (R Core Team 2025) with the `brms` package⁸³. Posterior samples were
898 generated in `rstan`¹⁰² using the No-U-Turn Sampling algorithm. Models were fit using 4
899 chains of 2000 iterations each with the first 1000 iterations from each chain discarded as
900 warmup. Posterior inference, plotting, and data wrangling relied largely on the packages
901 `tidyverse`, `tidybayes`, and `data.table`. Priors were determined using prior predictive simulation
902⁸¹. Model fit was assessed using posterior predictive checks¹⁰³.

903 **Modelling contaminant exports**

904 Using the posterior distributions of parameters from the water to adult concentration models
905 (see **Contaminant transfer to adult aquatic insects**), we estimated adult concentrations
906 for each hydrobasin based on the concentrations of a given compound predicted from the
907 random forest model (C_w ; see **Global modelling of aqueous contaminant concentration**),
908 such that:

909

$$910 \text{Conc}_{\text{basin}} = \alpha + \beta C_w$$

911

912 where $\text{Conc}_{\text{basin}}$ is the predicted tissue concentration (e.g., $\mu\text{g mgDM}^{-1}$) of a compound as a
913 function of the intercept α and slope β multiplied by the water concentration predicted from
914 the random contaminant model (C_w).

915 To estimate the export of each compound per hydrobasin, we multiplied $E_{i,\text{basin}} * \text{Conc}_{\text{basin}}$,
916 resulting in 919,156 estimates of the export of each compound. Summing across
917 hydrobasins (for each posterior iteration of $\text{Conc}_{\text{basin}}$) generated a distribution of estimates of
918 the global export of each compound from rivers and streams per year.

919 As shown above, contaminant export from each basin is a function of the total insect
920 emergence from that basin and the concentration of contaminants in an insect, which itself is
921 a function of the aqueous contaminant concentrations. Therefore, to estimate the relative
922 contribution of emergence biomass versus aqueous concentration to contaminant export by
923 insects, we first fit a linear model of the form $\text{Export} \sim \text{intercept} + \beta_1 E_{i,\text{basin}} + \beta_2 C_w$ using the
924 `lm()` function in R. The data came from 50,000 randomly sampled hydrobasins. We then
925 used the `calc.relaimpo()` function from the `relaimpo` package¹⁰⁴ to estimate the proportion
926 of total variance explained by each predictor variable.

927

928 **Data availability**

929 All contaminant data and geospatial data are publicly available and can be accessed via the
930 references provided in the Methods section. Data for all analyses are available at

931 <https://github.com/Aqua-Sync/secondary-production> (to be permanently archived at Zenodo
932 upon acceptance).

933

934 **Code availability**

935

936 Code and data are available at <https://github.com/Aqua-Sync/secondary-production> (to be
937 permanently archived at Zenodo upon acceptance).

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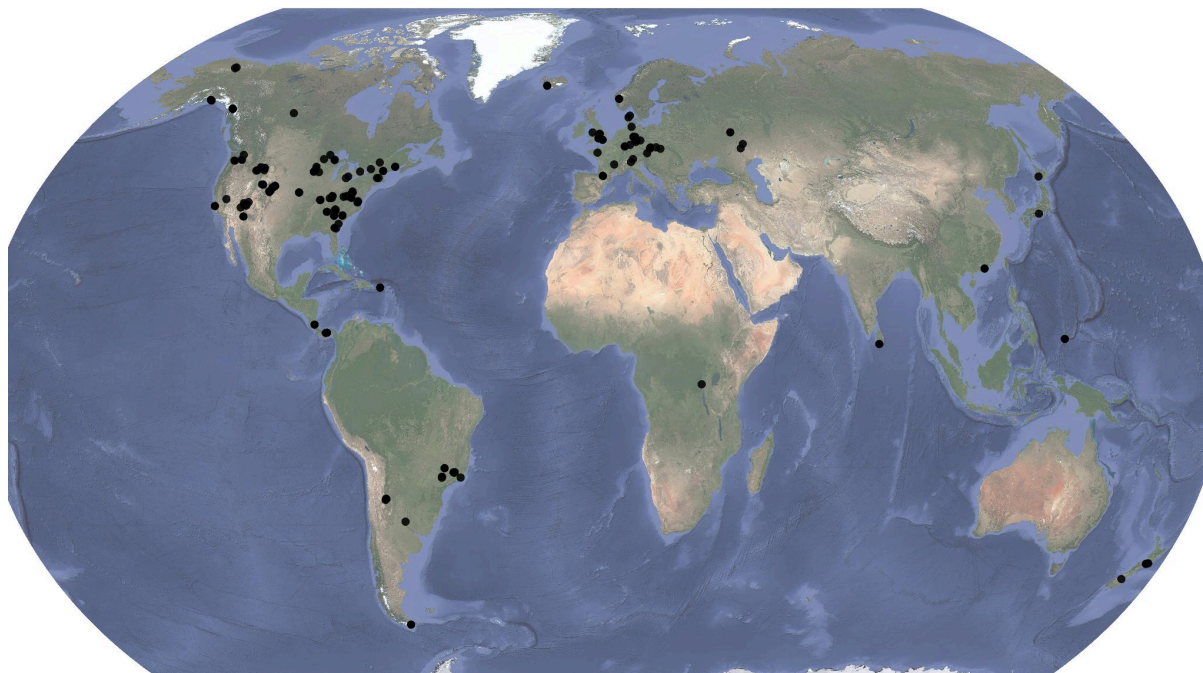
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1233 Extended DATA - Figures

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1236 **Extended Data Fig. 1-** Global map depicting the locations of the 305 reports used for secondary
1237 production estimates.

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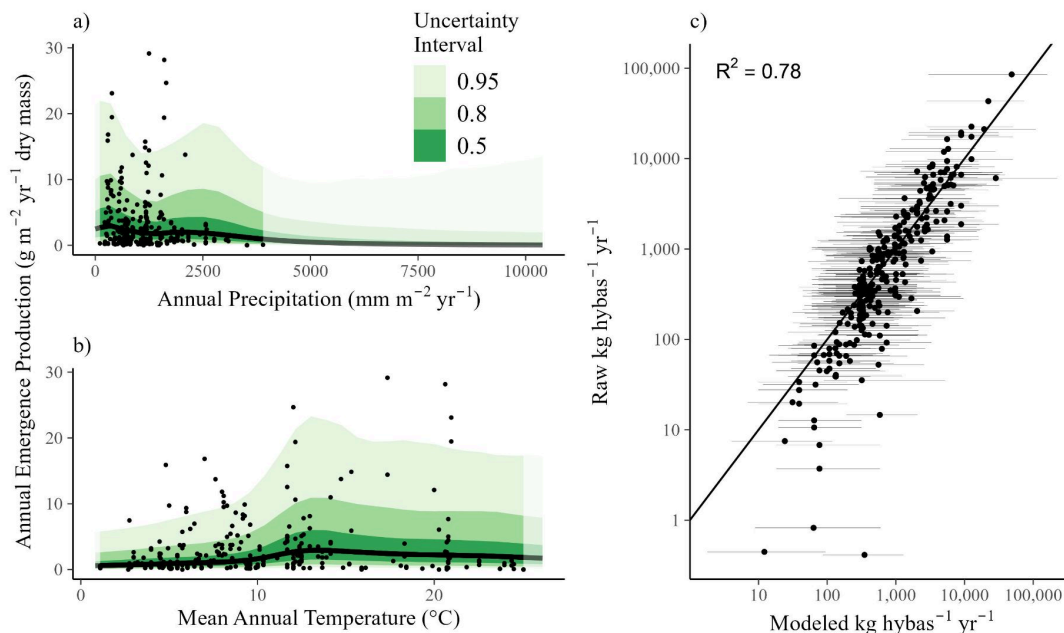
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1245 **Extended Data Fig. 2** a and b) Posterior predictions of annual insect emergence production fit to 305
 1246 samples (dots). The black line is the posterior median, with shading showing 50, 80, and 95% credible
 1247 (uncertainty) intervals (CrI). The transparency shifts denote predictions within the training data (dark)
 1248 versus outside the training data (light). Most precipitation values in the training data is <2500 mm m⁻²
 1249 y⁻¹ (a), whereas the temperature data cover nearly the entire global range (b). c) Modeled annual
 1250 insect emergence per hydrobasin versus observed (n = 305) annual emergence per hydrobasin. The
 1251 solid line shows the 1:1 relationship. Dots and error bars show the posterior median and 95% CrI (for
 1252 the modeled data) or the observed values for the empirical data.

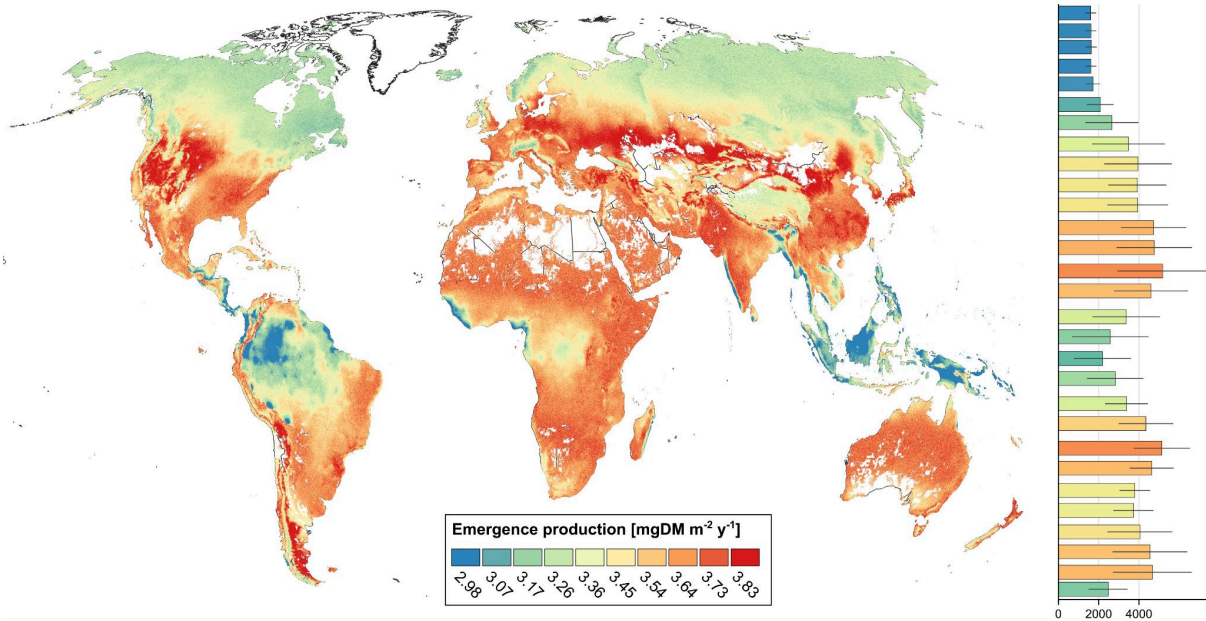
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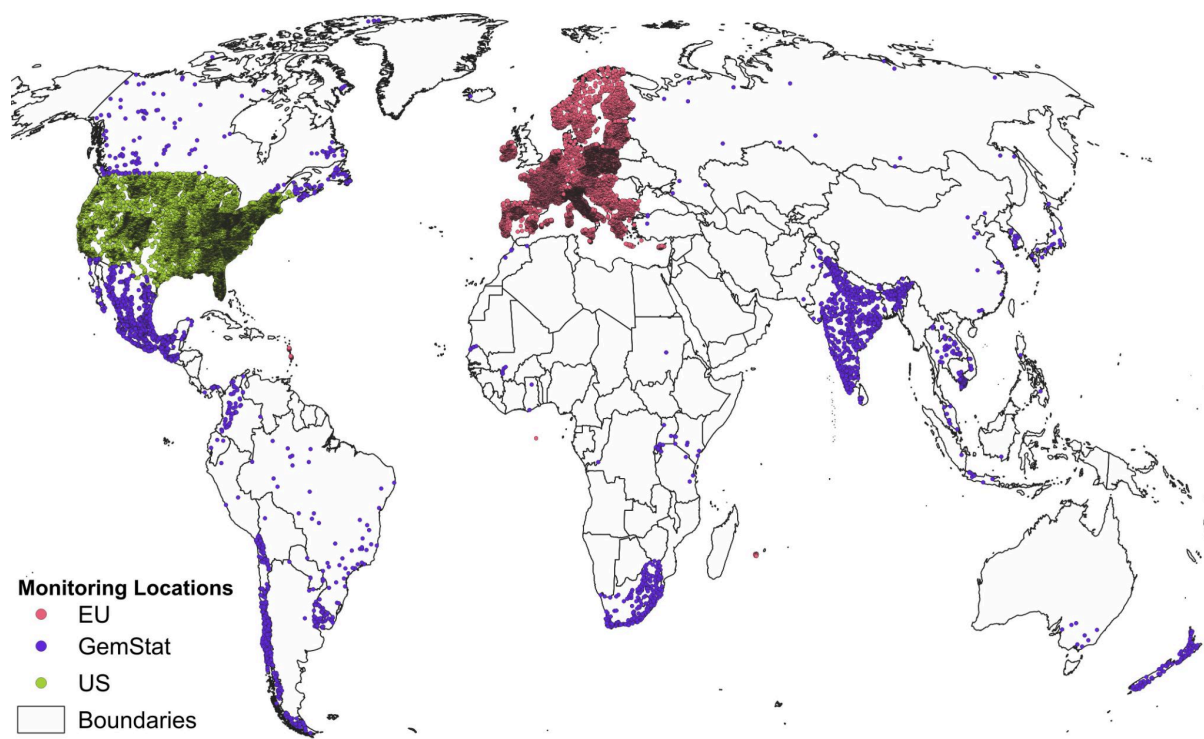
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Extended Data Fig.3 - Using predictions of the Bayesian hierarchical GAM model, we estimated aquatic insect emergence production (mgDM m⁻² y⁻¹) in each of the 919,156 sub-basins. Median production ranged from <1,100 mgDM m⁻² y⁻¹ in the Arctic (Greenland, Siberia, and North American arctic) to >1,500 mgDM m⁻² y⁻¹ in primarily temperate regions of Europe, North America, and Asia. Bar graphs depict the average ($\pm$ SD) insect emergence production (mgDM m⁻² y⁻¹) per 5° degree latitude bin.



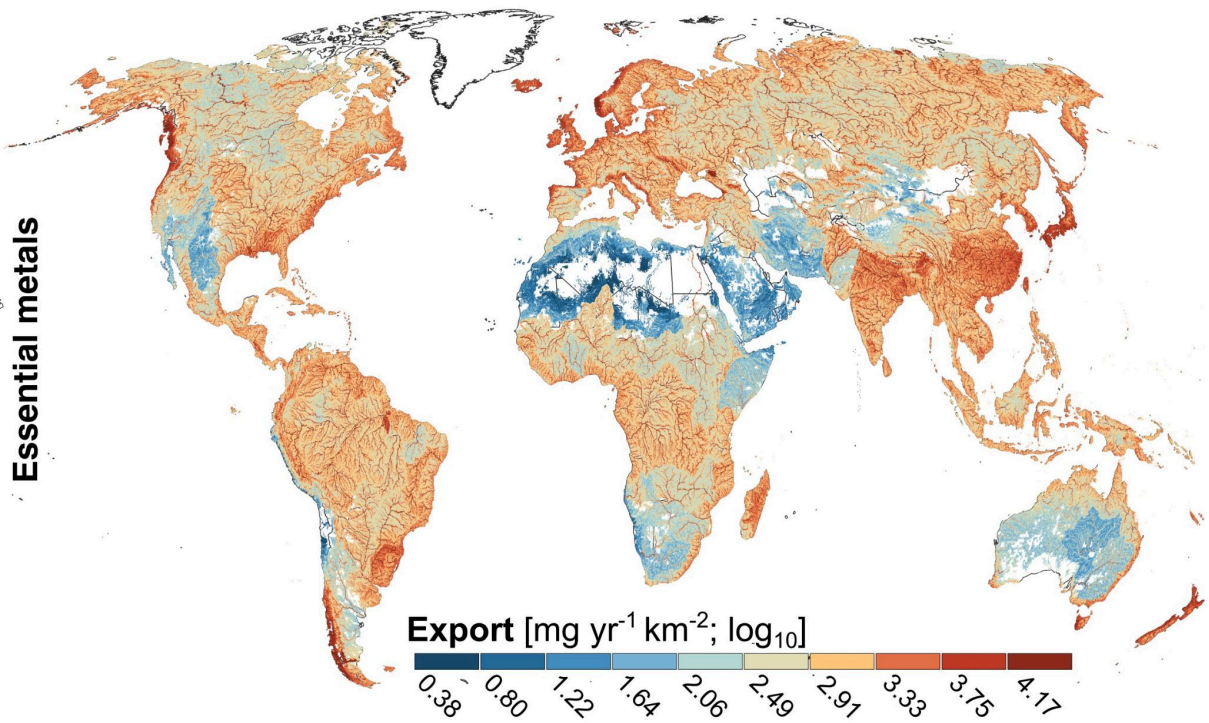
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1273 **Extended Data Fig. 4** - Location of contaminant concentration data used for model training (red =
 1274 Europe, Water Information System Europe [WISE]; green = US, Water Quality Portal [WQP]) and
 1275 validation (blue = global, Global Freshwater Quality Database [GEMStat]). Training data for the 37
 1276 contaminants were taken from the EU WISE ($n \approx 7.4$ million) and US WQP ($n \approx 5.3$ million), whereas
 1277 the global validation data were provided by the GEMStat database ($n \approx 3.2$ million) curated by the
 1278 German Federal Institute for Hydrology.

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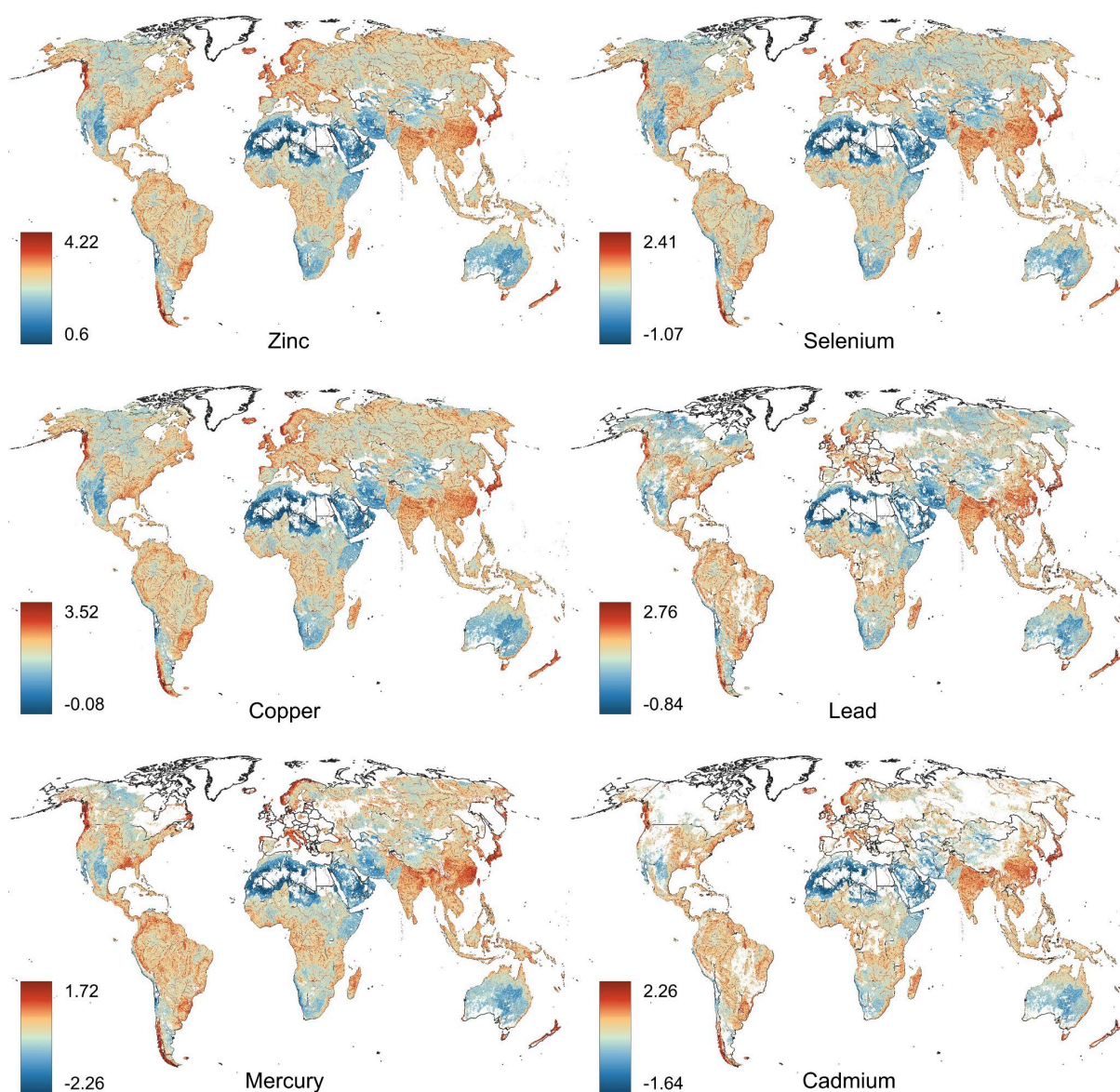


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1282 **Extended Data Fig. 5** - Global insect-mediated export of the sum of the three essential metals
 1283 (copper, zinc and selenium) depicted as rates in mg per year per square kilometer. Export rates are
 1284 $\log_{10}$ -transformed.

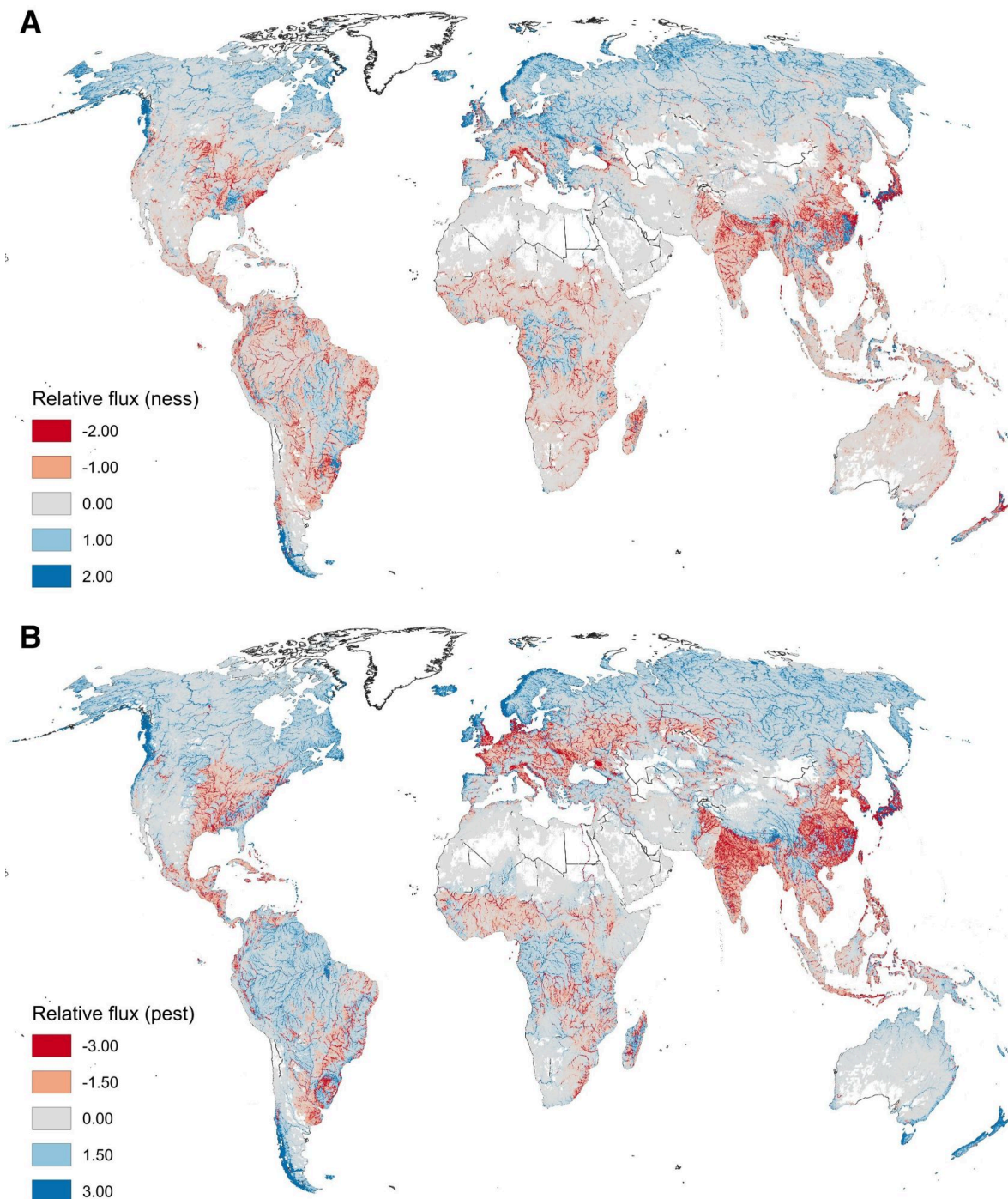
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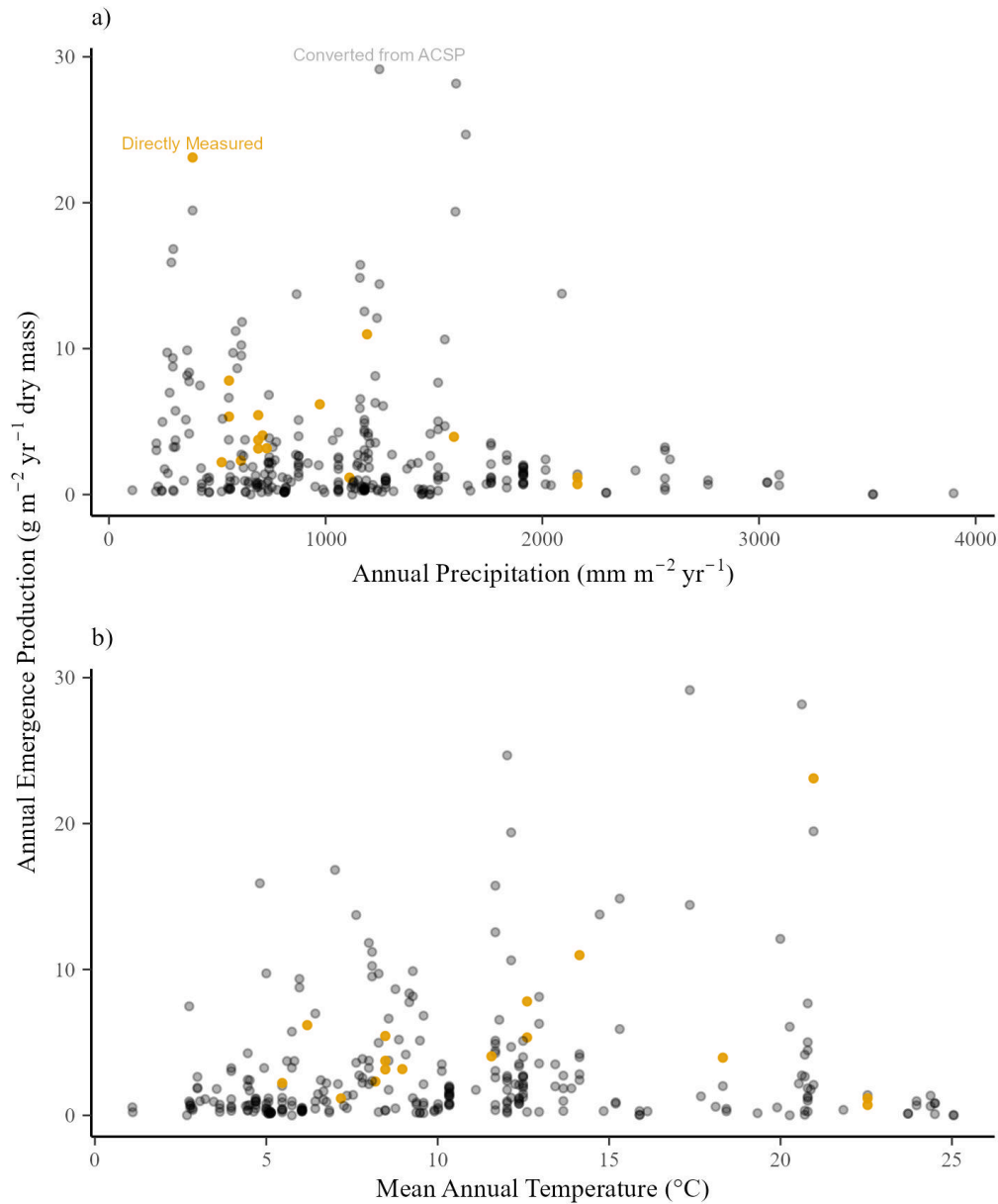
1288 **Extended Data Fig. 6** Global annual export rates in mg km^{-2} for three essential metals (zinc,
 1289 selenium, copper) and three non-essential metals (lead, mercury, cadmium). Export rates are
 1290 $\log_{10}$ -transformed for ease of interpretation; note that color scales vary across panes and thus cannot
 1291 be compared.



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1293 **Extended Data Fig. 7:** Global maps of relative export fluxes comparing beneficial (i.e.,
 1294 polyunsaturated fatty acids; PUFA) and potentially adverse (i.e., contaminants) export rates. Relative
 1295 export maps highlight the disbalance between beneficial and potentially adverse export rates with
 1296 respect to the global average export of both PUFAs and contaminants. Disbalances should not be
 1297 interpreted as higher environmental risks to consumer groups, but as indicators of larger potential
 1298 trade-offs for higher consumers. (A) Global map of relative export for non-essential metals (ness;
 1299 sum of Cd, Hg, and Pb). (B) Global map of relative export for pesticides (pest, sum of herbicides,
 1300 insecticides and fungicides). All relative export values are multiplied by 10^6 to improve readability.

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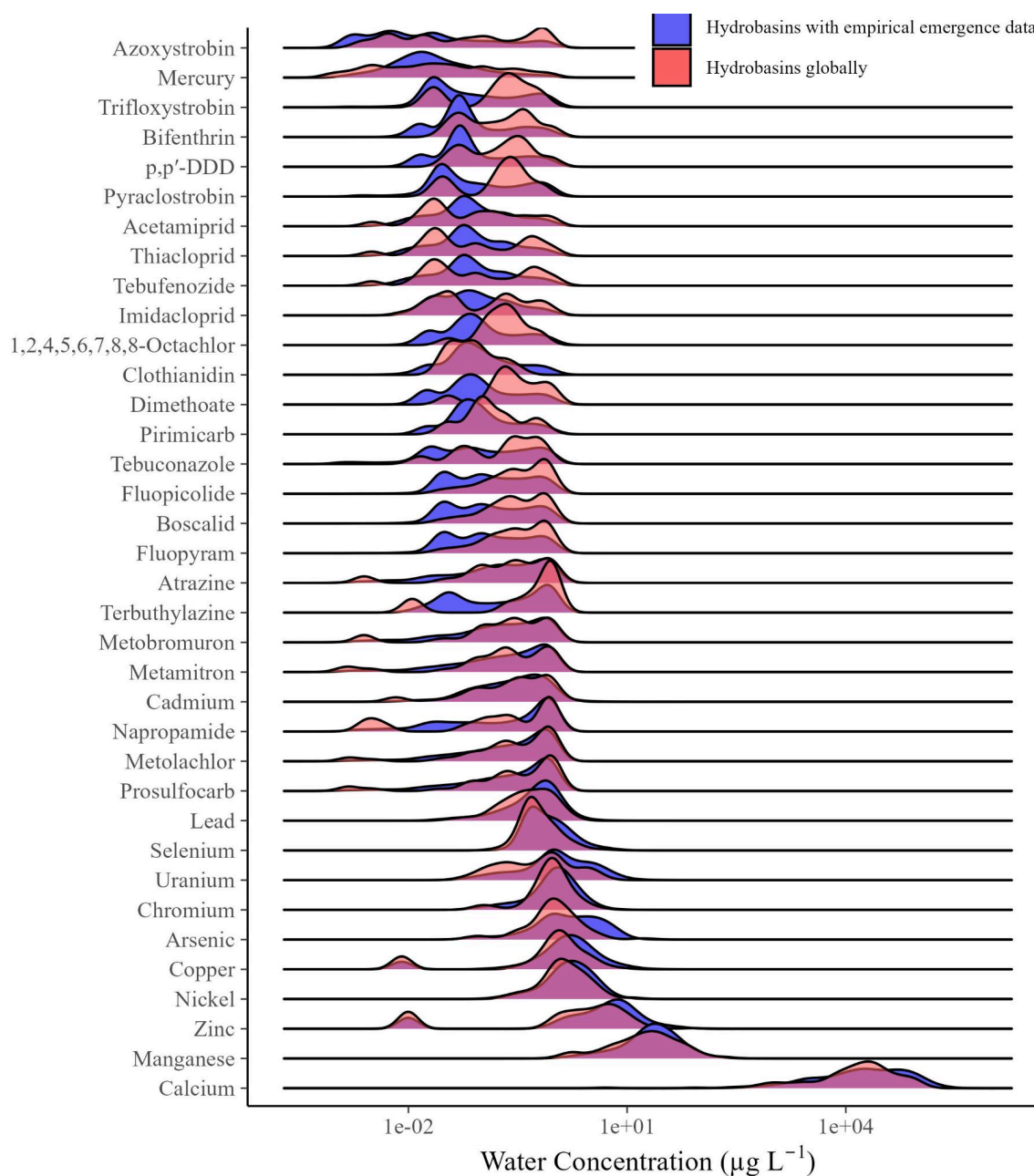
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1303 **Extended Data Fig. 8** - Comparisons of our modeled estimates of insect emergence (black dots) to
 1304 directly measured annual insect emergence (orange dots). Plots are shown for the full dataset (All
 1305 Data) and for data that were converted from annual community secondary production (converted from
 1306 ACSP) versus those measured directly from emergence studies (Directly Measured). Most extracted
 1307 production data reported ACSP, which is annual benthic production of all macroinvertebrates. As
 1308 described in the methods, we converted these measures to just adult insect emergence using
 1309 empirical relationships between the fraction of benthic production that is only insects and the fraction
 1310 of insect production that emerge successfully to adults. The similarity in distributions between
 1311 converted and measured values (i.e. overlap of the dots) indicates that our conversion procedure
 1312 produced biologically reasonable estimates of insect emergence.

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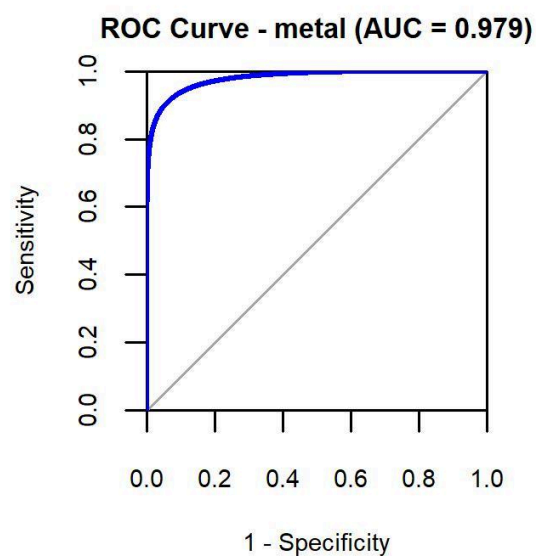
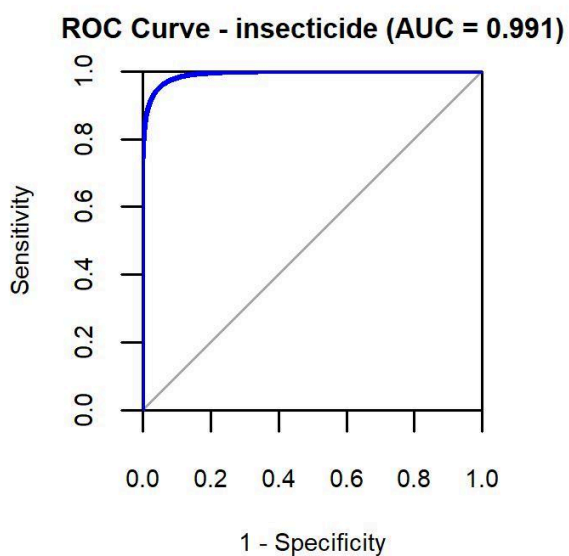
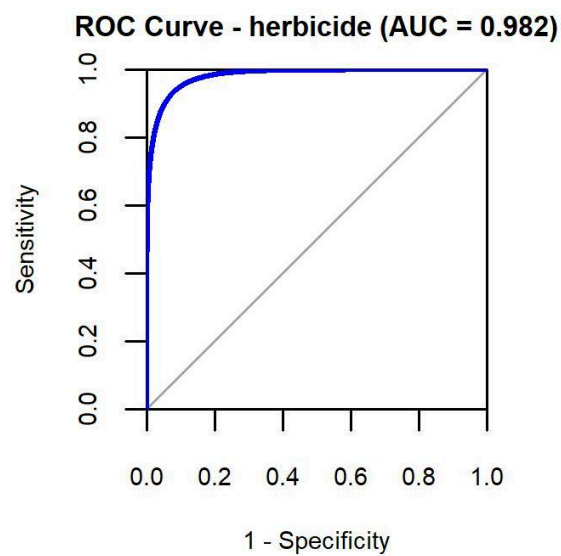
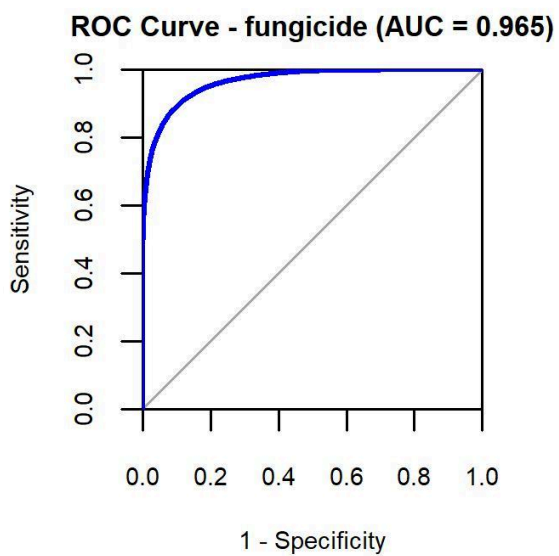
1317 **Extended Data Fig. 9** - Comparison of contaminant water concentrations globally (red) with
 1318 concentrations only in the hydrobasins that had direct estimates of secondary production or
 1319 emergence (blue). The strong overlap suggests that the hydrobasins used to sample emergence were
 1320 typical in terms of water quality to global distributions.

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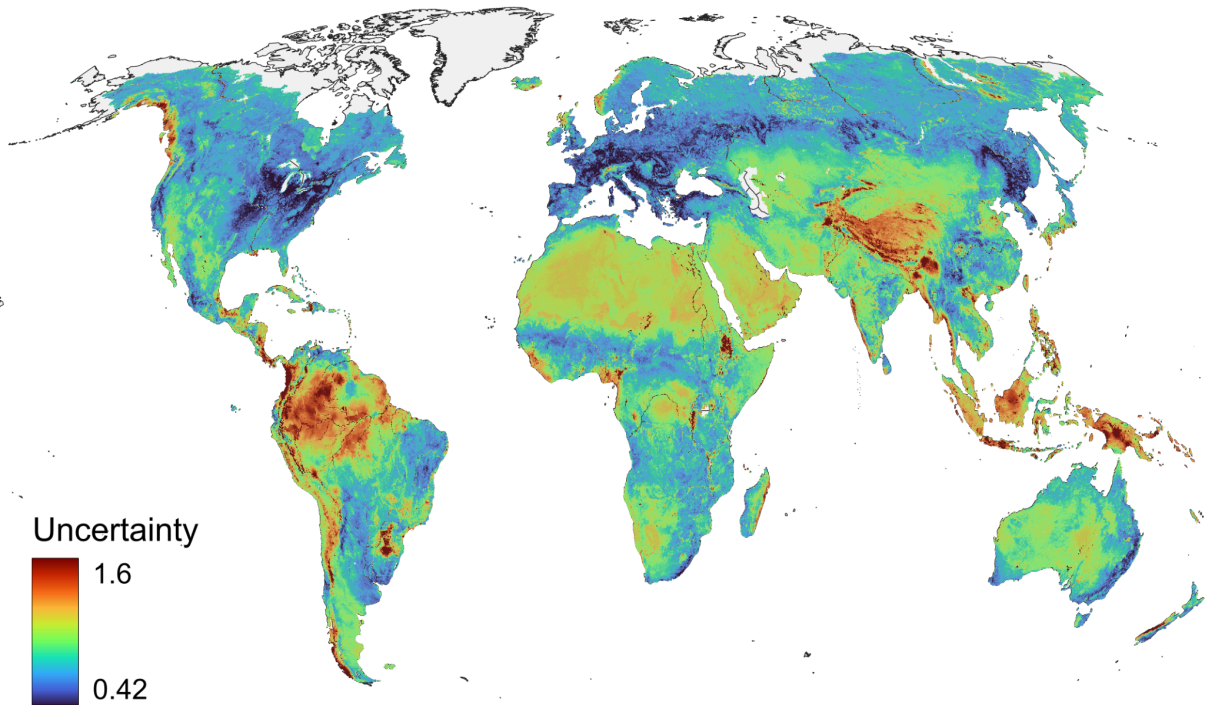
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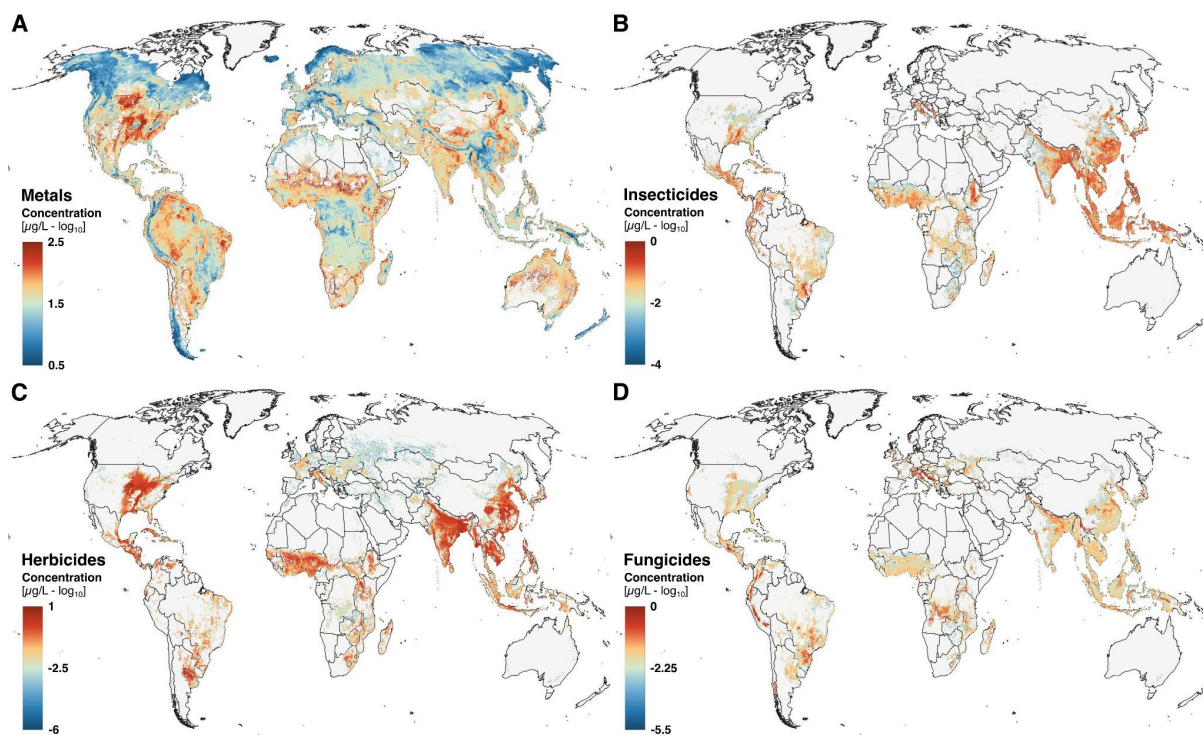
1326 **Extended Data Fig. 10** - Receiver-operator curves for the four classification models with their areas
 1327 under the curve (AUC) annotated at the top.



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1329 **Extended Data Fig. 11** - Uncertainty associated with global contaminant predictions derived for the
 1330 24 main numerical predictors. Uncertainty is depicted via the mean absolute deviance across all
 1331 predictors (expressed as z-score normalized parameters). Z-score normalization was conducted with
 1332 respect to the training data (Europe and the US) to highlight divergences of parameter matrices
 1333 outside the training area. For instance, in the western Amazonia basin (dark red), spatial predictors
 1334 are on average 1.6 standard deviations away from the the average predictor value used during
 1335 training, whereas in Central Europe (dark blue) spatial predictors are, on average, only 0.42 standard
 1336 deviations away from the parameter room used during training.

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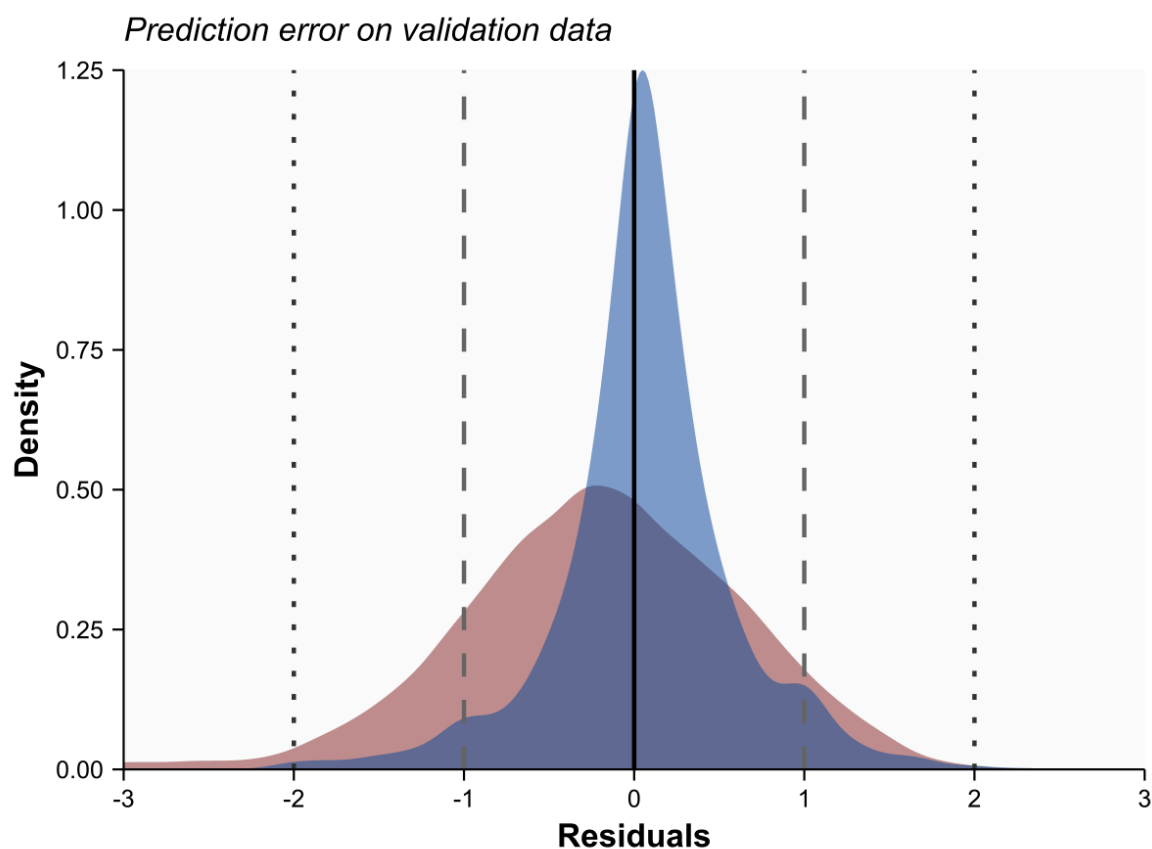


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1339 **Extended Data Fig. 12** - Global average water contaminant concentrations for (A) metals, (B)
 1340 insecticides, (C) herbicides and (D) fungicides displayed as log10-transformed concentrations in μg
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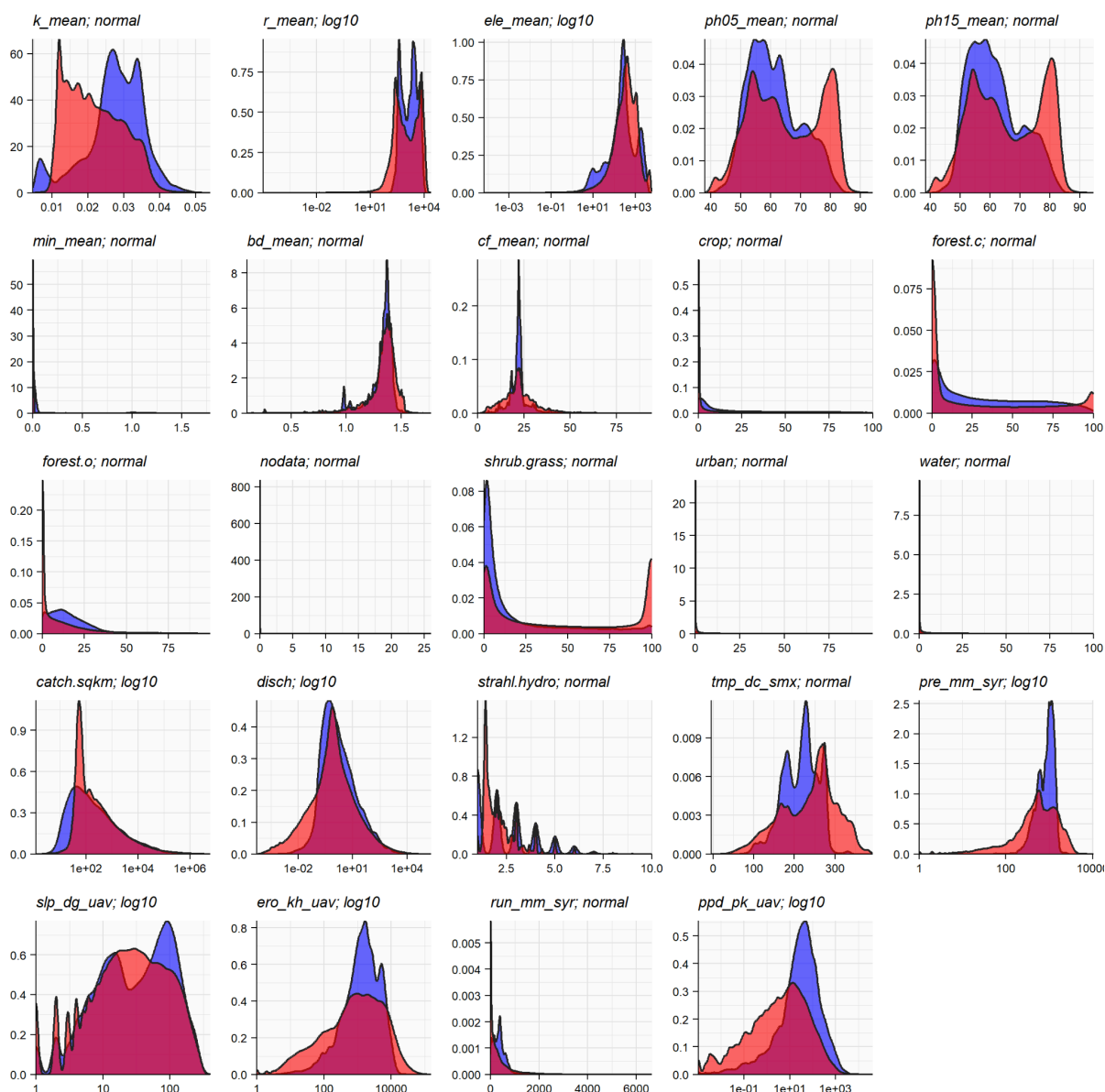


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1345 **Extended Data Fig. 13** - Error of concentration prediction models using GEMStat data as an
 1346 independent validation source. Errors are expressed as residuals ($\log_{10}$) of predicted vs. actual
 1347 measurements within the training area (i.e., Europe and US, blue) and outside the training area (i.e.,
 1348 rest of the world, red). Negative residual values represent underestimations of predicted
 1349 concentrations.

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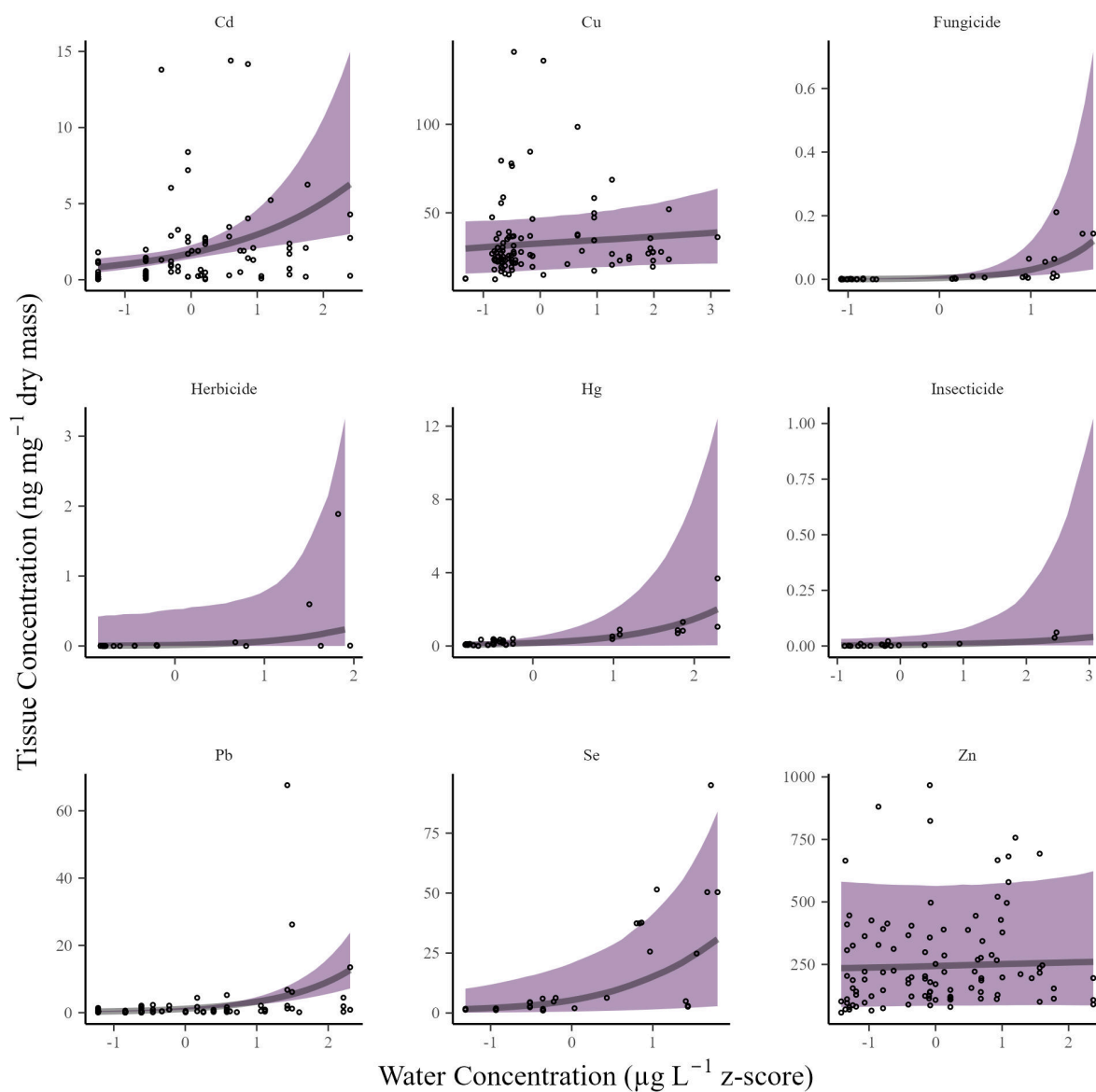
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1353 **Extended Data Fig. 14 - Density overview of numerical predictors for global contamination**
 1354 *modelling with blue density curves representing predictor distributions of locations used for*
 1355 *training and red distributions depicting all global sites for which contaminant models were*
 1356 *applied. Definitions of predictors are in Extended Table 4.*

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1360 **Extended Data Fig. 15** - Modeled relationships between tissue concentrations of contaminants in
 1361 adult aquatic insects and corresponding aqueous concentrations. Lines and shading show the
 1362 posterior median and 95% Crl. Dots show the raw data.

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1373 Extended Data Tables

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Extended Data Tab.1. Comparison of data used to fit regression models of emergence (Data in GAM Models) and the global range of data across all hydrobasins (Global Data). Summaries show the median, 95% confidence interval (CI), the range (minimum to maximum), and the total number of replicates (N).					
Predictor	Units	Data	Median (95% CI)	Range	N
Precipitation	mm m ⁻² y ⁻¹	Data in GAM Models	1096 (176.9 to 2792.5)	52 to 4084	305
Precipitation	mm m ⁻² y ⁻¹	Global Data	541 (15 to 2791)	0 to 10392	919,156
Stream Temperature	Mean annual °C	Data in GAM Models	9.2 (2.9 to 24.1)	1 to 25	305
Stream Temperature	Mean annual °C	Global Data	18.4 (1 to 25.3)	0.8 to 26.1	919,156

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Extended Data Tab. 2 - Relative importance of annual emergence biomass and aqueous concentrations in explaining variation in insect-mediated contaminant export from river and streams. Values show the contribution of a model's total R^2 that is explained by each predictor.

Contaminant	Relative Importance	
	Biomass Export	Aqueous Concentration
Pb	0.9	0.078
Se	0.96	0.016
Cu	0.98	0.0078
Hg	0.91	0.023
Zn	0.98	0.0031
Cd	0.97	0.0043
Herbicides	0.2	0.18
Insecticides	0.32	0.056
Fungicides	0.26	0.058

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Extended Data Tab. 3. Parameter definitions used in modelling the export of biomass, nutrients, and contaminants in adult aquatic insects from rivers and streams. For equations, “Statistical model” indicates that the parameter is derived from the fit of a regression model that is explained in the text.

Parameter	Definition	Units	Equation
μ_{f_i}	Mean fraction of total community production that is comprised only of insects		Statistical model
P_b	Annual benthic 2° production - all taxa	$mgDM \cdot m^{-2} \cdot y^{-1}$	
P_i	Annual benthic 2° production - insects only	$mgDM \cdot m^{-2} \cdot y^{-1}$	$P_i = P_b \mu_{f_i}$
E_i	Annual emergence production	$mgDM \cdot m^{-2} \cdot y^{-1}$	$E_i = \mu_{E:P} P_i$
$E_{i,basin}$	Annual emergence production per hydrobasin	$kgDM \cdot y^{-1}$	
$\mu_{E:P}$	Mean ratio of emergence to insect production		Statistical model
$E_{i,global}$	Annual global emergence production	$tDM \cdot y^{-1}$	$\sum E_{i,basin}$
C_w	Contaminant concentration in water		
C_a	Contaminant concentration in adult aquatic insects	$ng \cdot mgDM^{-1}$	
$C_{a,basin}$	Contaminant concentration in adult aquatic insects in each hydrobasin	$ng \cdot mgDM^{-1}$	Statistical model
$C_{f,basin}$	Contaminant export from each hydrobasin	$mg \text{ or } kg \cdot y^{-1}$	$C_{a,basin} \cdot E_{i,basin}$
$C_{a,global}$	Contaminant export globally		$\sum C_{f,basin}$
μ_F	Mean concentration of PUFAs (EPA + DHA) in adult aquatic insects	$ng \cdot mgDM^{-1}$	Statistical model
$Conc_{basin}$	Tissue concentration of a contaminant or nutrient in adult aquatic insects, predicted across basin		$Conc_{basin} = \alpha + \beta C$

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Extended Data Tab. 4. List of predictors used to model annual insect emergence production. All predictors were standardized to z-scores before modelling. See Extended Data Table 5 for the list of models.				
Variable	Description	Resolution	Source	Notes
precip	Mean annual precipitation	HYBAS 12	https://www.hydroshed.org/hydroatlas	Converted to precipitation in mm before analyzing
streamtemp	Mean annual stream temperature	HYBAS 12	https://www.hydroshed.org/hydroatlas	Converted from mean annual air temperature (tmp_dc_syr in HydroSheds)
hft_ix_s93	1993 Human Footprint Index (at sub-basin pour point)	HYBAS 12	https://www.hydroshed.org/hydroatlas	
hft_ix_u93	1993 Human Footprint Index (total watershed upstream of sub-basin pour point)	HYBAS 12	https://www.hydroshed.org/hydroatlas	
hft_ix_s09	2009 Human Footprint Index (at sub-basin pour point)	HYBAS 12	https://www.hydroshed.org/hydroatlas	
hft_ix_u09	2009 Human Footprint Index (total watershed upstream of sub-basin pour point)	HYBAS 12	https://www.hydroshed.org/hydroatlas	
ele_mt_sav	Elevation	HYBAS 12	https://www.hydroshed.org/hydroatlas	
dis_m3_pyr	Discharge	HYBAS 12	https://www.hydroshed.org/hydroatlas	log10 transformed before standardizing and modelling
for_pc_sse	Percent Forest Cover	HYBAS 12	https://www.hydroshed.org/hydroatlas	
crp_pc_sse	Percent Cropland	HYBAS 12	https://www.hydroshed.org/hydroatlas	

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Extended Data Tab. 5 - Model selection for predicting mean annual insect emergence (y) as a function of abiotic predictors. Predictors are indicated in parentheses of f(), where f() represents the spline function in a Generalized Additive Model (GAM) fit in `brms`. Definitions of predictors are in Extended Data Table 4. Intercepts include the global intercept (α) and varying intercepts (i.e., random effects) for author and year of publication (au) and for the hydrobasin ID (hy). Model selection results are shown as the expected log posterior density and its standard error (elpd $\pm$ se), where less negative numbers indicate better fit. To generate global predictions of flux, we used Model 3, which contains an interaction of the two most common predictors (precipitation and temperature) in the top suite of models.

Model #	Model	elpd $\pm$ se
1	$y \sim \alpha + f1(\text{precip}) + \text{au} + \text{hy}$	0 ± 0
2	$y \sim \alpha + f1(\text{streamtemp}) + \text{au} + \text{hy}$	-1.8 ± 2.4
3*	$y \sim \alpha + f1(\text{streamtemp}, \text{precip}) + \text{au} + \text{hy}$	-2.3 ± 3.3
4	$y \sim \alpha + \text{au} + \text{hy}$	-2.4 ± 2.7
5	$y \sim \alpha + f1(\text{streamtemp}) + f2(\text{precip}) + \text{au} + \text{hy}$	-2.9 ± 2.9
6	$y \sim \alpha + f1(\text{hft_ix_s93}) + \text{au} + \text{hy}$	-3.5 ± 2.3
7	$y \sim \alpha + f1(\text{hft_ix_u93}) + \text{au} + \text{hy}$	-3.6 ± 3.9
8	$y \sim \alpha + f1(\text{hft_ix_s09}) + \text{au} + \text{hy}$	-5.2 ± 3.7
9	$y \sim \alpha + f1(\text{hft_ix_u09}) + \text{au} + \text{hy}$	-5.3 ± 3.5
10	$y \sim \alpha + f1(\text{elemsav}) + \text{au} + \text{hy}$	-5.6 ± 4
11	$y \sim \alpha + f1(\log10(\text{dism3pyr})) + \text{au} + \text{hy}$	-6.7 ± 4.1
12	$y \sim \alpha + f1(\text{for_pc_sse}) + \text{au} + \text{hy}$	-7 ± 4.2
13	$y \sim \alpha + f1(\text{crp_pc_sse}) + \text{au} + \text{hy}$	-7.7 ± 4.1
14	$y \sim \alpha + f1(\text{streamtemp}) + f2(\text{crp_pc_sse}) + \text{au} + \text{hy}$	-8.6 ± 4.2
15	$y \sim \alpha + f1(\text{streamtemp}) + f2(\text{crp_pc_sse}) + \text{for_pc_sse} + \text{au} + \text{hy}$	-9.7 ± 5.5

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1412 **Extended Data Tab. 6** - Overview of spatial predictors attributed to monitoring data for
 1413 machine-learning models with their data-prefix, resolution and direct source. Land-use land cover
 1414 predictors (prefix h) encompass six spatial predictors (e.g. percent agriculture, percent urban areas)

Type	Prefix	Resolution	Source
Elevation (DEM)	ele	1 km	https://www.earthenv.org/topography
Bulk density topsoil	bd	0.03°	https://doi.org/10.1594/PANGAEA.868929
Clay fraction topsoil	cf	0.03°	https://doi.org/10.1594/PANGAEA.868929
Lithological map	lit	0.01°	https://doi.pangaea.de/10.1594/PANGAEA.788537
Land-use land cover	h	0.001°	https://zenodo.org/records/3939038
K factor (soil erodibility)	k	0.01°	https://esdac.jrc.ec.europa.eu/content/global-s-oil-erodibility
R factor (rainfall erosivity)	r	0.008333°	https://esdac.jrc.ec.europa.eu/content/global-rainfall-erosivity#tabs-0-description=1
Soil pH (5-15 cm)	pH05	0.008333°	https://files.isric.org/soilgrids/latest/data/
Soil pH (15-30 cm)	pH15	0.008333°	https://files.isric.org/soilgrids/latest/data/
Global Mining footprint	min	0.008333°	https://download.pangaea.de/dataset/942325/allfiles.zip https://zenodo.org/records/7894216
Upstream catchment	catch	HYBAS 12	https://www.hydrosheds.org/hydroatlas
Discharge	disch	HYBAS 12	https://www.hydrosheds.org/hydroatlas
Strahler order	strahl	HYBAS 12	https://www.hydrosheds.org/hydroatlas
Air temperature	tmp	HYBAS 12	https://www.hydrosheds.org/hydroatlas
Precipitation	pre	HYBAS 12	https://www.hydrosheds.org/hydroatlas
Slope	slp	HYBAS 12	https://www.hydrosheds.org/hydroatlas
Soil erosion	ero	HYBAS 12	https://www.hydrosheds.org/hydroatlas
Run-off	run	HYBAS 12	https://www.hydrosheds.org/hydroatlas
Population density	ppd	HYBAS 12	https://www.hydrosheds.org/hydroatlas

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Extended Data Tab. 7 - Accuracy metrics for occurrence prediction models. Values reported are error rate of the null model (majority class), error rate of the full training model (out-of-bag), error rate of the testing data (misclassification), precision, recall, F1-score, relative improvement (RI, in percent), and the total number of data (n) used for model building and testing.

Model	Error _{null}	Error _{train}	Error _{test}	Precision	Recall	F1-score	RI %	n
fungicides	19.7	7.7	7.74	0.93	0.98	0.95	60.9	247,217
herbicides	24.5	7.2	7.2	0.94	0.97	0.95	70.6	940,724
insecticides	8.6	2.4	2.4	0.98	1.00	0.99	72.1	900,707
metals	65.2	8.5	8.46	0.87	0.88	0.88	87.0	5,243,841

1426 **Extended Data Tab. 8** - Accuracy metrics for concentration prediction models. Values reported
 1427 are mean squared error (MSE, null and full model), relative improvement (RI, in percent),
 1428 mean absolute error (MAE, null model), R-squared (training and testing) and the total
 1429 number of data (n) used for model building and testing.

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Model	MSE _{null}	MSE _{full}	RI %	MAE _{null}	R ² train	R ² test	n
fungicides	0.448	0.080	82.1	0.478	0.823	0.821	48,722
herbicides	0.762	0.107	86.0	0.690	0.860	0.860	230,347
insecticides	0.774	0.066	91.4	0.625	0.911	0.914	77,143
metals	2.807	0.100	96.4	1.206	0.964	0.964	3,416,562

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